

ComplexGitSync 3.1.0

Documentation

Nicolas Flipo

September 23, 2026

Contents

1	Getting Started	7
1.1	Prerequisites	7
1.2	Installation	7
1.3	Create or Check a Project Spec	7
1.4	Initialise the Workspace	8
1.5	Inspect the Runtime State	9
1.6	Run the Git Cycle	9
1.7	Log Files	10
1.8	Next Steps	10
2	Direct Python Object API	11
2.1	Programmatic Project Configuration	11
2.2	From Empty Registry to READY via <code>.gts</code>	11
2.3	Object-Level Tag Propagation	12
2.4	READY Methods in the Client API	12
2.5	Landing a Branch: <code>merge</code> and <code>refresh_private</code>	14
2.6	Targeting a Single Path: <code>add</code> and <code>remove</code>	15
2.7	<code>.gitignore</code> Lifecycle Sync and Commit Identity	16
2.8	Forcing a Clone Protocol	16
2.9	Repository Discovery: <code>discover_repos</code>	17
2.10	Submodule Migration: <code>import_submodules</code>	17
2.11	Adopting a Submodule Tree: <code>init_from_submodules</code>	18
2.12	Looking at a memory: <code>memory_status</code> , <code>memory_list</code> , <code>memory_show</code> , <code>memory_explore</code> 18	
2.12.1	Starting a memory's history over: <code>memory_reboot</code>	20
2.13	Machine-readable output: <code>status_json</code> and <code>verify_json</code>	20
2.14	Environment observation and drift	21
2.15	A memory as a repository: <code>memory_init</code> , <code>memory_clone</code> , <code>memory_push</code>	21
2.16	Register Integrity: <code>verify</code>	22
3	Architecture Overview	23
3.1	Three-Tier Architecture	23
3.2	Architectural Positioning	23
3.3	Tier 1 — Core Data	24
3.4	Tier 2 — Actions	25
3.5	Tier 3 — Client / API	27
3.6	Module Layout by Tier	28
3.7	Document Hierarchy	29
3.8	Registry Model	29

4	Writing cgs Files: The Complete Authoring Guide	33
4.0.1	What is a cgs File?	33
4.0.2	Minimal cgs: The Simplified Form	33
4.0.3	Normalization: What the Simplified Form Expands To	34
4.0.4	Inline Table Syntax: Overriding Defaults	35
4.0.5	Advanced Table Syntax: Legacy Form	36
4.0.6	Repository Placement and Path Resolution	36
4.0.7	Nested Configuration Discovery	37
4.0.8	Access Protocols and Custom Providers	38
4.0.9	Branches: default_branch and fallback_branch	38
4.0.10	Private Repositories	39
4.0.11	Complete Reference: All Recognized Fields	39
4.0.12	Practical Examples	40
4.0.13	Authoring with CLI Tools	41
4.0.14	Validation and Common Errors	42
4.0.15	Templates and Starting Points	42
4.0.16	Best Practices	43
4.0.17	Internal Round-Trip Guarantee	43
5	User Guide	45
5.1	CLI Overview	45
5.1.1	Prerequisites	45
5.2	Lifecycle Contract	46
5.3	Commands	47
5.3.1	initialise	47
5.3.2	bootstrap	48
5.3.3	discover	49
5.3.4	configure	49
5.3.5	create-cgs	50
5.3.6	clean-init	50
5.3.7	purge	50
5.3.8	pull	50
5.3.9	pull-force	51
5.3.10	clone	51
5.3.11	checkout	52
5.3.12	close-branch	52
5.3.13	add	53
5.3.14	rm	53
5.3.15	commit	53
5.3.16	push	53
5.3.17	freeze-release	54
5.3.18	freeze	54
5.3.19	import-submodules	55
5.3.20	init-from-submodules	56
5.3.21	memory	56
5.3.22	verify	59
5.3.23	env and env check	60
5.3.24	launch-release	60
5.3.25	view-tree	61
5.3.26	What cgitsync promises a script	61
5.3.27	Where a command works: the default workspace	62
5.3.28	status	62

5.3.29	<code>merge</code>	64
5.3.30	<code>pull -private</code>	65
5.4	Document Formats	65
5.4.1	Local Authoring Spec (<code>.cgs</code>)	65
5.4.2	Branches in a <code>.cgs</code>	67
5.4.3	Repository Placement and Nested Discovery	70
5.4.4	Git Tree State Snapshot (<code>.gts</code>)	71
5.4.5	Local Git Register and Sync Ledger (<code>.lgr</code>)	72
5.5	Worked Examples: From <code>CGS111</code> to <code>cawaqsviz</code>	72
5.5.1	Level 1 — <code>CGS111</code> : the minimal synthetic topology	73
5.5.2	Level 2 — <code>cawaqs</code> : the same minimal style, at scale	74
5.5.3	Level 3 — <code>cawaqsviz</code> : three ways to the same <code>.cgs</code>	74
5.6	Configuration and Environment	76
5.7	Error Reference	76
5.8	Troubleshooting	76

Chapter 1

Getting Started

This chapter guides a new user through the Multi-repo Sync CLI. The documented entry point is `cgitsync`.

The primary lifecycle is:

1. `initialise(.cgs)` → clone all repositories → `READY`
2. `initialise(.gts)` → load or restore a saved snapshot → `READY`
3. `pull(.cgs|.gts)` → resynchronise an existing tree
4. `checkout / add / commit / push / tag` → tree-wide Git operations
5. `freeze` → write the next `.gts` snapshot and update the project-local `.lgr`

1.1 Prerequisites

- Python 3.11 or later.
- Git 2.30 or later available on `PATH`.
- Working Git authentication for every remote used by the workspace.

1.2 Installation

Install `ComplexGitSync` from source with the Pixi-managed environment:

```
$ git clone https://github.com/flipoyo/ComplexGitSync.git
$ cd ComplexGitSync
$ pixi install
$ pixi run cgitsync --help
```

1.3 Create or Check a Project Spec

A `.cgs` file is the local authoring spec for the repository tree. It normally contains only a project name and repository identifiers:

```
project = "CGSil1"
repos = [
  "gitlab:CGS_test/CGSil1",
  "gitlab:CGS_test/CGSil2",
  "github:flipoyo/CGSih1",
]
```

Each identifier uses `provider:owner/repository`. ComplexGitSync runs `PARSE -> NORMALIZE -> VALIDATE` and supplies deterministic defaults: `main` branches, `ssh`, `nested_config = auto`, and the repository name as its path. A unique repository matching the project name is mounted at `..`. Advanced tables are reserved for overrides.

Canonical providers are `github`, `gitlab`, `codeberg`, and `custom`. Their known hosts are `github.com`, `gitlab.com`, and `codeberg.org`; SSH and HTTPS remotes are generated automatically. Custom hosting requires an explicit `gitprovider_url` and never guesses a host. For example, `codeberg:GX4G/GX4G` identifies owner `GX4G` and repository `GX4G` on Codeberg.

Copy `examples/template.cgs` when starting a specification. It combines plain repository identifiers with one inline override. Developers can compare it with `examples/normalized_template.cgs`, which shows the complete canonical document after normalization and is not intended as the usual hand-authored form.

The common inline options are `default_branch`, `fallback_branch`, `access_protocol`, `relative_path`, and `nested_config`. A relative path is resolved from the parent repository. Nested config can be `auto` to load the sole root-level `*.cgs`, `disabled` to stop discovery, or a relative `.cgs` path to select an exact file inside the repository.

```
$ pixi run cgitSync validate install.cgs
$ pixi run cgitSync view-tree install.cgs
```

A valid declared tree prints a lifecycle line such as:

```
DECLARED ready=false complete=true
```

1.4 Initialise the Workspace

Use `initialise` as the canonical first command. From a `.cgs` file it clones the full repository tree, validates it, writes the runtime snapshot under `<workspace>/cgitSync/state/`, and leaves the tree `READY`.

```
$ pixi run cgitSync initialise install.cgs
```

The same canonical project definition can be authored directly. The `-repo` option is repeatable:

```
$ pixi run cgitSync initialise --project CGSil1 \
  --repo gitlab:CGS_test/CGSil1 \
  --repo codeberg:GX4G/GX4G
```

Use either a source file or `-project` plus `-repo`, never both. To serialize the CLI definition without initializing a workspace, use `create-cgs -project ... -repo ... -output FILE`.

When `-output-path` is omitted, `cgitSync` uses the workspace-level default `CGSPATH=../..`, then derives `CGSHOME=CGSPATH/<project-name>`. The explicit equivalent is:

```
$ pixi run cgitSync initialise install.cgs --output-path "$CGSPATH"
```

From a previously generated snapshot (resolved automatically via the project's `.lgr` register; pass the path explicitly only to override that discovery):

```
$ pixi run cgitSync initialise
```

Successful `.cgs` initialisation prints a readable operation sequence `GT-LOAD->GT-DISCOVER->GT-VALIDATE->` the explicit workflow `load->expand->validate->clone->gitignore`, the per-repository `git clone` action, the resolved root path, a report of any `.gitignore` changes, and a compact tree outline. Structured log records also put the operation code first, for example `GT-DISCOVER`, `GT-VALIDATE`, `FS-PURGE`, `GT-CLONE`, or `GT-GITIGNORE`. If a stale partial checkout blocks the

clone step, `initialise` fails explicitly and prints `Try clean-init method`. Use `clean-init` to insert a purge phase between validation and cloning:

```
$ pixi run cgitssync clean-init install.cgs
```

The explicit operation sequence is `GT-LOAD->GT-DISCOVER->GT-VALIDATE->FS-PURGE->GT-CLONE->GT-GIT`. The matching workflow is `load->expand->validate->purge->clone->gitignore`. The purge phase can also be run on its own:

```
$ pixi run cgitssync purge install.cgs
```

`purge` removes generated clone state from `CGSHOME`: immediate child repository directories declared directly under the project root, and project `*.lgr` files. A persisted Git identity override at `.cgitsync/master.toml` (see the `initialise` entry in the Command Reference chapter for `-git-user-name/-git-user-email`) is workspace configuration, not clone state, and is left untouched.

1.5 Inspect the Runtime State

```
$ pixi run cgitssync view-tree
$ pixi run cgitssync view-tree --depth 2 --collapse parent_2
$ pixi run cgitssync status
```

If a command accepts an optional snapshot and none is provided, `cgitssync` discovers the latest `.gts` automatically via the project's `.lgr` register under `CGSHOME/.cgitsync/`. Each generated `.gts` snapshot actually lives under its own content-addressed `CGSHOME/.cgitsync/state/` directory; pass an explicit path (e.g. `-gts FILE`, or a positional source, depending on the command) only to override that discovery. `CGSHOME` can be set explicitly; for `initialise(.cgs)` the default is `CGSPATH=../..`, and later commands can also discover the workspace by walking upward from the current directory.

1.6 Run the Git Cycle

Once the tree is `READY`, the Multi-repo Sync commands operate on every repository in deterministic order:

```
$ pixi run cgitssync pull
$ pixi run cgitssync checkout feature/my-branch
$ pixi run cgitssync add
$ pixi run cgitssync commit "feat: update project CGS#1"
$ pixi run cgitssync push
$ pixi run cgitssync freeze-release release-2026.05 "release 2026.05"
$ pixi run cgitssync freeze release-2026.05
$ pixi run cgitssync launch-release release-2026.05
```

`pull` walks the tree parent-first, including the project root: `root -> parent -> leaf`. Mutation commands such as `add`, `commit`, `push`, and `freeze` walk the opposite direction: `leaf -> parent -> root`. When local files block the safe pull, the CLI suggests `cgitssync pull-force`; that recovery command is destructive and discards local uncommitted/untracked work before force-updating the same tree.

Pass `-gts <snapshot.gts>` when you want to pin a command to an explicit snapshot, and pass `-dry-run` to `add`, `commit`, `push`, or `freeze` to preview the plan without mutating repositories.

1.7 Log Files

Every CLI command writes a timestamped log file. On Linux the default location is `/.local/state/ComplexG`. If `XDG_STATE_HOME` is set, logs are written under that state directory instead. The CLI prints the resolved path as `log_file=<path>`.

1.8 Next Steps

See Chapter 5 for the command reference, document formats, preflight checks, and troubleshooting notes.

Chapter 2

Direct Python Object API

This chapter shows how to manipulate `ComplexGitSync` objects directly (without the CLI wrapper), including loading a `.gts` snapshot, reconstructing the runtime registry, mirroring identity into `GitTree/GitRepo`, and propagating a tag target.

For normal Multi-repo Sync usage, prefer the CLI lifecycle documented in the README and Getting Started guide. The reference test topology is CGSill (https://gitlab.com/CGS_test/CGSill): `initialise -> pull -> checkout/add/commit/push -> freeze -> launch_release`.

The lower-level object/API lifecycle remains: `load(.cgs) -> .gts LOADED -> expand(.cgs|.gts) -> .gts PENDING -> validate(.cgs|.gts) -> .gts READY -> pull(.cgs|.gts) -> checkout(.gts) -> add -> git(tree,"commit",msg) -> git(tree,"push") -> git(tree,"tag",name) -> freeze`.

2.1 Programmatic Project Configuration

Project configuration does not require the CLI or an existing `.cgs` file. The public facade accepts authoring values non-interactively and delegates normalization, static validation, and optional serialization to `cgs_format.py`. It performs no Git or network operations.

```
from ComplexGitSync import ComplexGitSyncClient

client = ComplexGitSyncClient()
document = client.configure(
    "GX4G",
    ["codeberg:GX4G/GX4G"],
    output_path="GX4G.cgs", # optional
)
reference_tree = document.to_git_tree()
```

The CLI commands `configure`, `create-cgs`, and `direct initialise -project/-repo` authoring are adapters over this method.

2.2 From Empty Registry to READY via `.gts`

```
from pathlib import Path

from ComplexGitSync import GitRepo, GitTree, RefKind, TreeLifecycleState
from ComplexGitSync.orchestration import GtsDocument,
    build_registry_from_gts_document
from ComplexGitSync.git_tree import WorkingGitTree

# 1) Empty registry lifecycle
empty_registry = WorkingGitTree()
```

```

assert empty_registry.recompute_tree_state() == TreeLifecycleState.
    UNLOADED

# 2) Load the CGSil1 runtime snapshot (.gts) and rebuild the dependency
    registry
cgshome = Path("../..")
gts_doc = GtsDocument.from_toml(cgshome / ".cgitsync/state/CGSil1.gts")
registry = build_registry_from_gts_document(gts_doc)
assert registry.lifecycle_state == TreeLifecycleState.READY

# 3) Rebuild GitTree/GitRepo identity objects from discovered registry
    entries
tree = GitTree()
for entry in registry.values():
    tree.add_repo(
        GitRepo(
            project_owner_name=entry.project_owner_name or "owner",
            project_name=entry.project_name or entry.name,
            gitprovider=entry.gitprovider,
            access_protocol=entry.access_protocol,
            commit_sha=entry.commit_sha,
        )
    )

```

2.3 Object-Level Tag Propagation

```

# Set the same tag target across all registry entries (parent -> leaves)
tree.propagate_tag(registry, "v1.2.3")
for entry in registry.values():
    assert entry.target_ref_kind == RefKind.TAG
    assert entry.target_ref_name == "v1.2.3"

```

This object-level flow is the same state model used by the CLI and `ComplexGitSyncClient`, but exposed directly for advanced automation and tests.

2.4 READY Methods in the Client API

The same lifecycle model is exposed by `ComplexGitSyncClient`. In practice, the workflow is:

- `load(...)` is the step-1 name for direct source loading.
- `expand(...)` is the canonical step-2 name; it runs nested `.cgs` discovery from parents to leaves (recursive).
- `validate(...)` is the canonical step-3 name; it accepts both `.cgs` and `.gts` sources.
- `initialise(...)` bootstraps a tree and must finish in `READY`; `clone(...)` is the direct clone entry point for a `.cgs` source; `pull(...)` resynchronises an existing tree.
- `git(tree, command, *args)` is the unified step-5 interface for "commit", "push", and "tag" operations. Each follows leaf-first ordering.
- `freeze(...)` emits the next persisted `.gts` state.

```

from pathlib import Path

from ComplexGitSync import ComplexGitSyncClient

client = ComplexGitSyncClient()
cgspath = Path("../..")
cgshome = cgspath / "CGSil1"

# 1. load the CGSil1 test-case .cgs -> .gts LOADED
client.load("../CGSil1.cgs")

# 2. expand(.cgs/.gts) -> .gts PENDING
print(client.expand("../CGSil1.cgs"))

# 3. validate(.cgs/.gts) -> .gts READY
client.validate("../CGSil1.cgs")

# 4. clone(.cgs/.gts)
# Same default as the CLI: output_path defaults to CGSPATH=../..
client.initialise("../CGSil1.cgs")

# Equivalent explicit form:
client.initialise("../CGSil1.cgs", output_path=cgspath)

# Recovery path for partial/stale clone state:
client.clean_initialise_cgs("../CGSil1.cgs", output_path=cgspath)

# Cleanup only:
client.purge_cgs("../CGSil1.cgs", output_path=cgspath)

# 5. READY methods - unified git interface
registry = client.get_dependency_registry()
# Pull uses ROOT -> PARENT -> LEAF; every repository in the tree is a
  plain
# independent clone (there are no gitlinks) and receives its own git
  pull.
# A .gts source must be an actual snapshot path (find the current one
  via
# RuntimeStateStore.latest_snapshot_for(), the same lookup the CLI uses
  when
# a command's snapshot argument is omitted) -- unlike a .cgs source,
  there is
# no fixed, predictable filename to hard-code here.
client.pull("../CGSil1.cgs")
# pull() fetches every branch of each remote, so checkout() can join a
  branch
# somebody else pushed: it starts the local branch from the remote-
  tracking
# ref when one exists, and only otherwise from the current HEAD.
client.checkout("feature/my-branch")
# Mutations use LEAF -> PARENT -> ROOT.
client.add()
client.git(registry, "commit", "feat: update CGSil1 CGS#1")
client.git(registry, "push")
client.git(registry, "tag", "v1.2.3")

# 6. freeze -> next .gts id

```

```
client.freeze("release-2026.05")
```

Every one of these that reaches a remote folds the memory first. When this tree mounts and has adopted a memory (`.cgitsync/.memory`), `push()`, `tag()`, and `freeze()` each call `memory_push` on it before doing their own work — regardless of `private`, since the memory is this project’s own record of itself, not a configuration repository scoped by that flag. `client.last_memory_fold` holds what the most recent one of these folded: the `memory_push` result dict, or `None` when no memory is mounted or the fold could not proceed. A fold that fails — the memory declared but not yet adopted, or unreachable for any other reason — is never raised: it is turned into a `warnings.warn(...)` call instead (catchable with `pytest.warns/warnings.catch_warnings`, the same as `checkout`’s own build-change warning), so a Python caller sees it exactly as a CLI user would, and the write it was folding for still completes. `client.memory_declared()` answers whether a memory is mounted at all, with no side effect, for a caller that wants to know in advance.

`close_branch(branch_name, private=False)`. Renames `branch_name` to `closed/branch_name` across the full tree, leaf-first — locally and on the remote, never deleting anything. Requires `READY`. Raises `GitSyncError` before touching any repository when `branch_name` is the project’s own default branch, or when any repository in scope is currently checked out on it — the error names which repository. `client.last_write_outcomes` holds one `RepoOutcome` per repository visited afterward: `acted` is `False` for a repository that had no branch of that name at all, never an error. `private=True` selects the writable configuration repositories instead of the project’s own, the same as `branch()` (`create`).

```
client.close_branch("memory-dev")
for outcome in client.last_write_outcomes:
    print(outcome.name, outcome.acted, outcome.detail)
```

2.5 Landing a Branch: `merge` and `refresh_private`

`merge(project_branch, private=False, ff_only=False, no_ff=False)` merges a branch across the tree, leaf-first, and returns one (`repo_name`, `merged_ref`) pair per repository a merge moved.

The argument is always the **project’s** branch name. Each repository resolves what that means for itself, so a private/local configuration repository merges `<its default_branch>_<project_branch>` rather than a branch of the project’s name, which it does not have. `merge_plan(project_branch, private=False)` returns one (`repo_name`, `source_ref`, `status`, `conflicting_paths`) row per repository without merging anything — what the CLI’s `-dry-run` prints. `status` is `"merge"`, `"already-on-it"`, `"no-branch"` or `"conflicts"`, and the last one carries the files that block it.

Every repository in scope is checked before any is merged, so a conflict anywhere leaves the whole tree untouched. The check names the conflicting files, and counts binary files: a tracked PDF changed on both sides conflicts just as a source file does.

`merge_resolve(project_branch, private=False, ff_only=False, no_ff=False)` is the opt-in alternative. It merges one repository at a time, stops at the first that conflicts, and returns a `ResolveOutcome` naming what it merged, where it stopped, the files that stopped it, and the repositories it never reached. It gives up the all-or-nothing guarantee on purpose: that is the only way to reach the conflicted worktree a merge tool needs.

`open_merge_tool(repo_name)` then opens one repository’s conflicts in a merge tool. It returns `None` once the tool has run, or the command to run by hand when no tool is available —

a missing editor is a normal outcome, not an error. A `merge.tool` you configured always wins, and your Git configuration is never written.

`refresh_private()` is the other direction, and what `pull -private` runs: it fetches and merges each private/local repository's base branch into the derived branch it is on, so a settings branch does not drift behind the project's while a feature branch is open.

```

client.merge("multi-branch")           # the project's own repos
client.merge("multi-branch", private=True) # ComplexGitSync_multi-
    branch                               #
                                         # -> ComplexGitSync
client.merge("multi-branch", all_writable=True) # both halves, one
    pass                                 #

# Merging into a branch that is not checked out, in one call.
client.merge_into_plan("multi-branch", "main") # what it would do, per
    repo                                       #
client.merge_into("multi-branch", "main")      # checkout and merge,
    together                                 #
client.build_installed_from("main")            # the version that
    branch carries                          #
client.refresh_private()                       # the other way round

```

all_writable: the API half of `-all`. `add`, `commit`, `push`, `merge`, `merge_resolve` and `merge_plan` each take `all_writable=False` alongside `private=False`. Setting it runs the call across this project's own repositories *and* its writable configuration ones in a single pass. The two are mutually exclusive and passing both raises; leaving both unset keeps today's behaviour exactly. Read-only configuration repositories are never written to under any combination — the parameter maps to `RepoScope.WRITABLE`, which is narrower than `RepoScope.ALL`.

2.6 Targeting a Single Path: add and remove

`add(paths=None)` and `remove(paths)` accept one or more filesystem paths (each relative to `CGSHOME`, or absolute) instead of operating on the whole tree:

```

# Stage only these two files, each in the repo that owns it -- every
# other repo in the tree is untouched.
client.add(paths=["notes.md", "deps/leaf/src/main.py"])

# Remove a single tracked file: deleted from disk, removal staged.
client.remove(["deps/leaf/obsolete.txt"])

```

Both resolve each path to its owning repo through the same helper, `resolve_repo_for_path(tree, path)` (`git_tree.py`) — the deepest (most specific) repo whose `absolute_path` contains the resolved path wins, so a nested child is correctly preferred over its parent. A path outside every repo in the tree raises `GitSyncError` immediately, before anything is staged or removed; for `remove`, a path that resolves to a directory or does not exist raises the same way rather than partially applying. `add()` with no `paths` keeps today's exact behaviour: every repo staged in full, leaf-first.

`remove` is a plain `git rm` (deletes from disk, stages the removal) — unrelated to `GitRunner.rm_cached` (index-only, used internally by `import_submodules` to drop a gitlink while keeping the child's working tree); the two are not interchangeable.

2.7 .gitignore Lifecycle Sync and Commit Identity

`initialise_cgs`, `initialise_cgs_document`, `clean_initialise_cgs`, `clean_init`, `restart`, and `pull` (for `.cgs` sources) all accept the same four optional keyword arguments, mirroring the CLI's `-commit-gitignore`, `-force-gitignore-sync`, `-git-user-name`, and `-git-user-email` flags:

```
from ComplexGitSync import ComplexGitSyncClient, MasterConfig

client = ComplexGitSyncClient()
client.initialise_cgs(
    "../CGSill.cgs",
    output_path=cgspath,
    commit_gitignore=True,          # stage/commit/push each changed .
                                   # gitignore
    force_gitignore_sync=False,    # pull-force fallback for a blocked
                                   # repo
    git_user_name="cgitsync-bot",
    git_user_email="bot@example.com",
)

# The identity above is persisted to CGSHOME/.cgitsync/master.toml, so a
# later call on the same workspace picks it up without repeating it:
client.initialise_cgs("../CGSill.cgs", output_path=cgspath,
    commit_gitignore=True)
assert MasterConfig.resolve_identity(cgshome, client.git_runner) == (
    "cgitsync-bot",
    "bot@example.com",
)
```

All four keyword arguments default to `False/None`: by default, every repository with children (root, or any nested repository that itself has further nested children) is safely pulled and has its `.gitignore` updated, but nothing is staged, committed, or pushed, and the commit identity — were a commit ever made — would be whatever `git config user.name/user.email` already resolves to locally. `MasterConfig` (`master.py`) is the class behind the identity override: `configure()` sets it in memory for the current process, `persist(cgshome, ...)` writes it to `CGSHOME/.cgitsync/master.toml` (and also updates the in-memory value), `load(cgshome)` reads a previously persisted override back in, and `resolve_identity(repo_path, git_runner)` returns the `(user_name, user_email)` pair — or `(None, None)` to defer entirely to local git config. This file is workspace-local configuration, not part of the `.cgs/.gts` project spec, and is preserved (not deleted) by `purge/clean-init`.

2.8 Forcing a Clone Protocol

`initialise_cgs`, `initialise_cgs_document`, `clean_initialise_cgs`, `clean_init`, `bootstrap`, and `clone_cgs` additionally accept `force_access_protocol`, mirroring the CLI's `-force-protocol` flag on `initialise`, `bootstrap`, and `clean-init` (`restart/pull` do not accept it):

```
# Every repo this run clones -- root siblings and any child discovered
# later from a nested .cgs in a different repo -- is cloned over https,
# regardless of what its own .cgs entry says. No .cgs file is read
# differently or written.
client.initialise_cgs(
    "../CGSill.cgs",
    output_path=cgspath,
    force_access_protocol="https",
```

```
)
```

"ssh" or "https"; None (the default) leaves every entry's own `.cgs`-declared `access_protocol` untouched. The override lives only in memory for the duration of the call – it is applied at the single point every clone's remote URL is built (`ComplexGitSyncClient._build_remote_url`), so it reaches root siblings and nested-discovered children alike without `cgs_format.py` parsing, or any `.cgs` file on disk, ever being touched differently.

2.9 Repository Discovery: `discover_repos`

`discover_repos(root_dir=None, *, max_depth=5, output=None)` scans a directory for git repositories already checked out on disk and drafts a `.cgs` from what it finds — the entry point for adopting a project that exists but has no specification yet:

```
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()

# Read-only scan - nothing is cloned, fetched, or modified
report = client.discover_repos("/path/to/checkout")
for repo in report.repos:
    print(repo.relative_path, repo.identifier, repo.branch)

for warning in report.warnings:
    print("unresolved:", warning)

# Write the draft out (same call, with an output path)
report = client.discover_repos("/path/to/checkout", output="draft.cgs")
```

The method returns a `DiscoverReport` dataclass with fields `root` (the scanned directory), `repos` (a tuple of `DiscoveredRepo`), `cgs_entries` (authoring-form repo tables ready for `configure()`), `warnings`, and `project_name`. Each `DiscoveredRepo` carries `relative_path`, `absolute_path`, `remote_url`, `identifier`, `branch`, and `has_cgs`.

Repository addresses are resolved through the existing `parse_repo_id()` grammar; a repository with no `origin` or an unrecognised provider is reported in `warnings` and omitted from `cgs_entries` rather than guessed at. Passing `output` raises `GitSyncError` when nothing resolvable was found, so an empty draft is never written.

2.10 Submodule Migration: `import_submodules`

`import_submodules(repo_root, *, apply=False, recursive=False)` converts git submodules in a local repository to plain `ComplexGitSync` nested clones:

```
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()

# Dry run - inspect without changing anything
report = client.import_submodules("/path/to/repo")
for sub in report.submodules:
    print(sub.name, sub.path, sub.url)

# Apply - convert every level, root first
report = client.import_submodules(
    "/path/to/repo",
    apply=True,
```

```

        recursive=True,
    )
    print(report.converted)    # tuple of converted submodule names

```

The method returns an `ImportSubmodulesReport` dataclass with fields `submodules` (all entries found), `applied` (whether the conversion ran), `converted` (names of converted submodules), and `scan_root` (the repository the call was pointed at, which every reported path can be expressed from). When `apply=False` (the default), no files are modified. When `apply=True`, each submodule is removed from the index (`git rm -cached`), its stanza is removed from `.gitmodules`, and the parent's `.gitignore` is updated to ignore the child path. The staged changes should then be reviewed and committed manually.

`recursive=True` also converts any submodule that has its own checked-out `.gitmodules`, at any depth. That walk follows the submodule graph declared by `repo_root`'s own `.gitmodules`, so a repository whose holder has already been converted is no longer reachable — which is why the conversion of a whole tree happens once, and last (see `init_from_submodules` below).

2.11 Adopting a Submodule Tree: `init_from_submodules`

`init_from_submodules(repo_root, *, cgs_path=None, max_depth=5, dry_run=False, force=False, force_access_protocol=None)` performs the whole adoption of a submodule-based checkout, in the one order that works:

```

from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()

# Report the plan without writing, cloning, or converting anything
plan = client.init_from_submodules("/work/project", dry_run=True)

# Adopt: discover -> write .cgs -> initialise -> convert every submodule
report = client.init_from_submodules("/work/project")
print(report.cgs_path)                # the .cgs used
print(report.tree.is_ready())         # True
print(report.import_report.converted) # converted submodule names

```

Each step delegates to the method that already owns it (`discover_repos`, `initialise_cgs`, `import_submodules`); the value this method adds is the ordering and the preconditions. `initialise_cgs` adopts the root in place but deletes and re-clones every other repository from its remote, where the submodules are still declared — so converting first is undone for every non-root repository, and a repair pass cannot reach any deeper level once the root's `.gitmodules` is gone.

The method returns an `InitFromSubmodulesReport` with fields `root` (which is also `CGSHOME`), `discover` (the `DiscoverReport` from step 1), `cgs_path`, `cgs_written`, `import_report`, `tree`, and `dry_run`. Nothing is committed: the conversion touches every repository that held a submodule, so `branch`, `checkout`, `add`, and `commit` stay explicit, separate calls. A `GitSyncError` is raised before anything is written when the discovery resolved nothing, when the project name does not match the directory name (`CGSHOME` is `<parent>/<project-name>`), or when `repo_root` has no `.gitmodules` to convert (`force=True` overrides the last one).

2.12 Looking at a memory: `memory_status`, `memory_list`, `memory_show`, `memory_explore`

The four read-only questions the CLI's `memory` group asks, each returning plain data rather than text:

```

from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()

status = client.memory_status("/path/to/CGSHOME")
status["states"], status["entries"]      # how much is remembered
status["verification"] # verified / no-history / legacy / corrupt /
    time-inconsistent
status["latest_toolchain"]["git"]        # what produced the newest
    record

for row in client.memory_list("/path/to/CGSHOME"):
    row["state"], row["recorded_at"], row["commands"]

state = client.memory_show("/path/to/CGSHOME", "2acdc98b") # hash or
    prefix
state["repos"], state["entries"]
for entry in state["entries"]:
    for commit in entry["commits"]:
        commit["message"], commit["repository"], commit["published"]
state["published"] # remote and ref each commit
    went to

answer = client.memory_explore("/path/to/CGSHOME")
answer["branch"] # the memory branch checked out
    here, or None
for commit in answer["commits"]: # published commits, newest
    push first
    commit["repository"], commit["branch"], commit["sha"], commit["
        message"]

timeline = client.memory_explore("/path/to/CGSHOME", timeline=True)
for entry in timeline["entries"]: # every ledger entry, in order
    entry["seq"], entry["command"], entry["commits"], entry["published"]

```

Commit messages come back whole. Shortening a long one to a single line is the caller's decision, which is why the CLI has `-full` and this method does not: a caller reading the memory from Python wants the message that was written, not the one that fitted.

`memory_show` raises `GitSyncError` when the prefix matches no State, or more than one.

`memory_explore`'s default view answers what a colleague pulling this branch would see — commits with no publication row are left out, because that is `memory_show`'s question, not this one. `branch=` asks for a memory branch other than the one checked out on this disk; today only the checked-out one can be read this way, and a mismatch raises `GitSyncError` naming `memory_clone(branch=...)` rather than answering for the wrong branch.

The commands that give a memory a repository of its own have the same mirror. `add_memory_repo_cgs` is the one the owner named; the CLI spells it `memory mount`.

```

client.repo_create("github:you/.memory", private=True)
# {"created": "created" / "exists" / "unavailable", "command": "gh
    repo create ..."}

client.add_memory_repo_cgs("project.cgs") # one entry, comments
    kept
client.memory_adopt("/path/to/CGSHOME") # .cgitsync/.memory,
    fresh and empty
client.memory_push("/path/to/CGSHOME") # folds .cgitsync's
    pending content in first

```

```

client.memory_branch("/path/to/CGSHOME", "main") # the branch a merge
    will need

# Only for a memory mounted before this command existed, directly at
# .cgitsync instead of .cgitsync/.memory:
client.memory_migrate("/path/to/CGSHOME", "project.cgs")

```

`repo_create` never raises for a repository that already exists — that is "exists", an ordinary answer — and never for a provider tool that is absent or signed out, which is "unavailable" carrying the command to run. It raises only for a provider no tool is known for, and for a tool that failed for some other reason. `memory_list` includes a State that entries name but that is not on disk, with `path` set to `None` — `verify` reports it as `MISSING_STATE`, and this is how a caller finds out which one.

2.12.1 Starting a memory's history over: `memory_reboot`

```

answer = client.memory_reboot("/path/to/CGSHOME")
answer["folded"]           # pending records folded in before archiving
answer["archived_from"]    # the branch's name before the reboot
answer["archived_to"]      # "<branch>.archived-<YYYYMMDD>", real, on
    origin
answer["exported"]         # .cgitsync/.memory/.cgs/<project>-v<N>.cgs
answer["branch"]           # the original name, fresh, empty, checked out

# A fresh start from the very first adopt, instead of inheriting
    whatever
# the fallback branch already holds:
client.memory_adopt("/path/to/CGSHOME", remote=remote, branch=branch,
    reboot=True)

```

Nothing is ever force-pushed or deleted: the old branch is pushed to origin under its archived name *before* the old name is removed from origin, so the commits it holds are reachable under some name on the remote for the whole operation, never only locally. `memory_reboot` raises `GitSyncError` when the mount is not a repository yet, and when today's archived name already exists — a second reboot the same day needs a name of its own rather than colliding with the first.

2.13 Machine-readable output: `status_json` and `verify_json`

Two methods return the same answers as `status()` and `verify()`, serialised as one JSON object each. They are what the CLI's `-json` flag prints, so a Python caller and a shell pipeline read exactly the same fields.

```

import json
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()
client.load_gts("/path/to/snapshot.gts")

payload = json.loads(client.status_json())
payload["cgitsync_branch"]           # the branch the tree is on
payload["summary"]["repos"]          # how many repositories it holds
payload["repositories"][0]["sync"]   # per repository, as the table shows
    it

chain = json.loads(client.verify_json("/path/to/CGSHOME"))

```

```
chain["status"]                                     # "clean" or "findings"
```

`verify_json` keeps the report it built on `client.last_verify_report`, so a caller needing both the object and the verdict does not verify the chain twice — which under `repair=True` would mean repairing twice.

The shape of both objects lives in `json_render.py`, defined once for every command rather than per command, and carries a `schema_version`. The promise attached to it is **additive only**: new fields may appear; existing fields do not change meaning and do not vanish.

2.14 Environment observation and drift

The client exposes the same read-only Environment operations as the CLI:

```
from ComplexGitSync.cgs_format import CgsDocument
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()
client.load_gts("/path/to/CGSHOME/.cgitsync/state/<hash>.gts")

observed = client.environment()
print(observed.architecture, observed.pixi_platform, observed.digest())

requirements = CgsDocument.from_toml("/path/to/project.cgs")
drift = client.check_environment(requirements)
if not drift.matches:
    print("missing", drift.missing)
    print("older", drift.older)
print("observed but undeclared", drift.undeclared)
```

`environment()` returns a secret-free, content-addressable `TreeEnvironment`. Credentials are booleans only; manifests are tree-relative paths and digests, never copied contents. When a source `.cgs` is available, `check_environment()` may be called without an argument. It returns data and installs nothing.

2.15 A memory as a repository: `memory_init`, `memory_clone`, `memory_push`

```
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()
client.load_gts("/path/to/CGSHOME/.cgitsync/state/<hash>.gts")

proposal = client.memory_init("/path/to/CGSHOME")
proposal["entry"]           # the .cgs entry that mounts the memory
proposal["branch"]          # this project's branch of the memory
    repository
proposal["create_with"]     # the command that creates it -- never run for
    you

client.memory_push("/path/to/CGSHOME")           # commit +
    push
client.memory_clone("/other/workspace", owner="flipoyo") # onto a new
    machine
```

`memory_clone` needs no loaded project — that is the case it exists for — so give it an `owner` and a `branch` when nothing is loaded, or a `remote` address directly for a memory that does not live where the convention says. It raises `GitSyncError` rather than writing over a memory already present, or when the branch it wants has never been pushed.

Calling `memory_push` by hand, as above, is for settling the memory on its own. *READY Methods in the Client API*, earlier in this chapter, covers the more common case: `push`, `tag`, and `freeze` each call it automatically, first, once a memory is mounted and adopted.

2.16 Register Integrity: `verify`

`verify(cgshome, *, repair=False)` checks the hash-chained `.cgitsync/lgr` register under `cgshome` for tamper-evidence:

```
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()
report = client.verify("/path/to/CGSHOME")

if report.is_clean:
    print("register is clean")
else:
    for seq, finding, detail in report.findings:
        print(seq, finding.name, detail)

# Repair a stale HEAD cache (never rewrites or deletes an entry)
client.verify("/path/to/CGSHOME", repair=True)
```

Returns an `integrity.VerificationReport`. Read `report.state` first: it is one of four `HistoryState` members — `VERIFIED`, `NO_HISTORY`, `LEGACY`, `CORRUPT` — and `report.is_verified` is `True` only for the first.

`report.is_clean` is a narrower question: it means “no findings were recorded”, which an empty register also satisfies. Use it to ask whether anything is *wrong*; use `state` to ask what was actually established. `findings` holds `(seq, Finding, detail)` tuples.

Every command that writes a `State` now appends an entry to `.cgitsync/lgr/`, so a workspace used since that landed answers `VERIFIED`. A workspace whose only history predates it answers `LEGACY`: the single-file register is still read, and nothing writes any more. `Finding` enumerates `BROKEN_LINK`, `BAD_ENTRY_HASH`, `SEQ_GAP`, `SEQ_DUPLICATE`, and `HEAD_STALE` for what this method actually checks (chain linkage and the `HEAD` cache); its remaining three members (`MISSING_STATE`, `ORPHAN_STATE`, `STATE_DIGEST_MISMATCH`) are reserved for a future store-level verification pass that cross-references entries against the actual `state/<hash>.gts` files on disk.

With `repair=True`, a stale `HEAD` cache is corrected in place. Register entries are never rewritten or deleted by this method, repair or not — a broken chain is reported, not silently healed.

Chapter 3

Architecture Overview

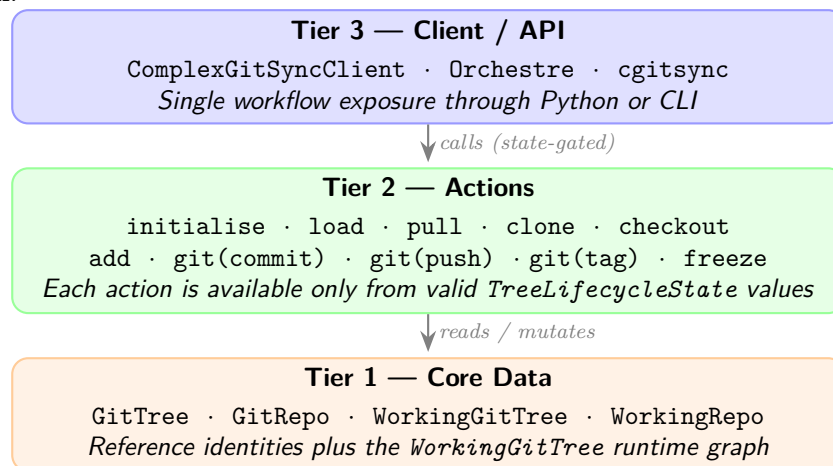
This chapter describes how `ComplexGitSync` is structured around three explicit tiers and explains the interaction between its classes. Related domain classes are grouped in modules according to the responsibility boundaries below.

3.1 Three-Tier Architecture

`ComplexGitSync` is organised into three tiers. Dependencies only flow **downward**: the Client calls Actions; Actions read and mutate Core Data; Core Data has no upward dependency.

Figure 3.1 shows the overall layout.

Figure 3.1: Three-tier architecture: Client/API exposes state-gated methods; Actions operate on Core Data; Core Data separates reference identities from the `WorkingGitTree` parent-leaf runtime graph.



3.2 Architectural Positioning

`ComplexGitSync` is positioned as a deterministic synchronisation layer *on top of Git*, not as a replacement for Git itself. It combines two distinct graph scopes:

- **Git DAG**: per-repository commit history and native remote semantics.
- **GitTree DAG**: workspace-level dependency graph coordinating parent/root/leaf propagation and ordering.

The system therefore differs from:

- plain Git usage (single repository scope),
- monorepo layouts (single storage boundary),
- submodule-only management (linking primitive without lifecycle orchestration),
- generic distributed sync engines (not constrained by Git-native ‘gts’/‘lgr’ deterministic contracts).

Figure 3.2 summarizes this positioning.

Figure 3.2: Architectural positioning matrix across Git scope and workspace coordination guarantees.

Approach	Primary scope	What it does not guarantee
Git (single repo)	Commit DAG per repository	Multi-repository propagation and coordinated lifecycle gating
Monorepo	One repository, one commit DAG	Independent repository identities and per-repo remote ownership
Submodule management	Parent repo references child revisions	Deterministic tree-wide mutation workflow and snapshot contracts
ComplexGitSync	Git DAG + GitTree DAG	Not a credential broker or remote policy authority

freeze materializes deterministic checkpoints as **.gts** snapshots (schema + SHA-256 hash), while **.lgr** assigns stable local ids and stores append-only ledger events for replayable local evolution.

Cross-declared nested repositories may initially resemble a local tangle at declaration time; runtime expansion normalizes this into a DAG via **fix_circularities**.

3.3 Tier 1 — Core Data

The reference model is centred on **GitTree**, which owns canonical **GitRepo** identities. Its runtime counterpart, **WorkingGitTree**, adds the parent–leaf graph of mutable **WorkingRepo** nodes. A project consists of one *root* repository that may nest *parent* repositories and terminal *leaf* repositories.

Figure 3.3 shows a representative project tree.

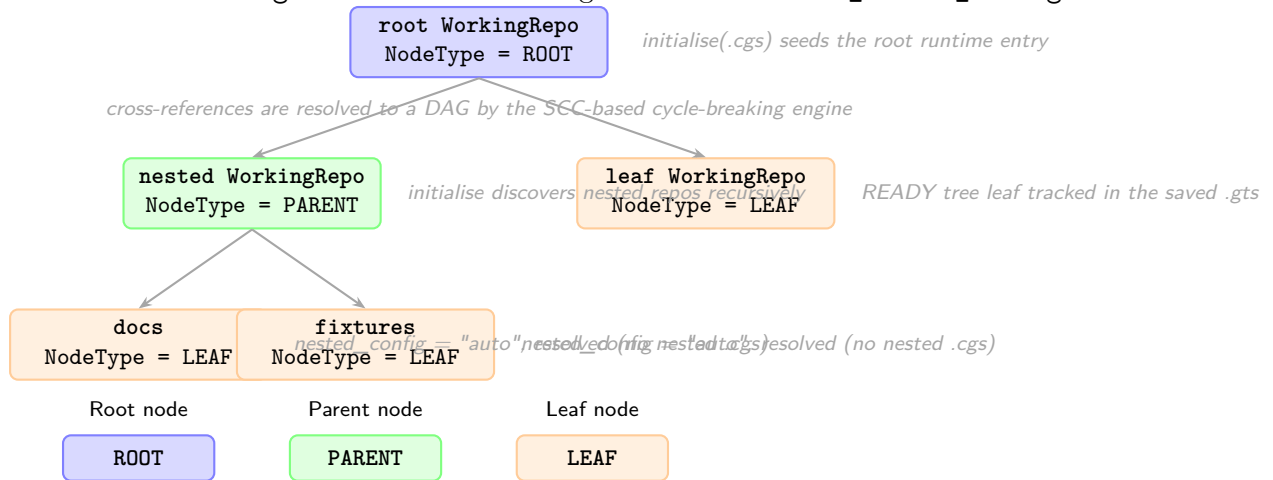
Registry entries in **WorkingGitTree** are indexed by a colon-separated repo ID derived from their position in the tree (e.g. **root:deps/child-repo:docs**). Cross-referenced parents therefore create a tangle-like declaration layer, but the expanded runtime graph remains a DAG after the cycle-breaking engine has run.

Cycle Breaking Engine

fix_circularities is a two-phase engine that converts a potentially cyclic dependency graph into a valid DAG:

1. **SCC detection (Tarjan’s algorithm)**. A directed dependency graph is built from the registry (edges: parent absolute path → child absolute path). **find_strongly_connected_components** identifies groups of paths that form a cycle. For each non-trivial SCC, an *Anchor* path is selected using three heuristics: (1) most external incoming edges, (2) closest to project root, (3) smallest SHA-256 hash. Back-edge entries — registry entries resolving to the anchor’s path but parented inside a non-anchor SCC node — are flagged **is_external_reference=True** and removed.

Figure 3.3: `WorkingGitTree` is a parent-leaf graph of `WorkingRepo` nodes. Root and parent nodes own nested `.cgs` files discovered during clone or `discover_nested_configs`.



2. **Hash-compatibility deduplication.** Residual path-duplicate entries are removed when their synchronisation state is compatible with the canonical entry.

The clone engine uses a **sync stack** (set of absolute paths already entered into the clone pipeline) to exclude entries whose path is already being processed, providing defence-in-depth against any residual back-references that escape the registry cleanup phase.

`topological_sort(registry)` returns entries in parent-first order (Kahn's BFS algorithm) for safe sequential clone/pull operations. Runtime `pull` uses the same three-level direction: `ROOT` → `PARENT` → `LEAF`; every repository — root, parent, and leaf alike — is a plain independent clone and is pulled in that same parent-first order, rather than through any submodule linkage.

Figure 3.6 shows the class overview across all three tiers, `GitTree/GitRepo` among them; the full current module-by-module Ring-level dependency graph lives in `docs/DevGuide/architecture.md`.

3.4 Tier 2 — Actions

Actions are operations that read or mutate the Core tier. **Every action is gated by the current `TreeLifecycleState`:** actions that require a `READY` tree will refuse to execute and log a gating-refusal event if the tree is not ready.

Figure 3.7 shows the full lifecycle state machine. The public CLI presents this lifecycle through `initialise`, `pull`, and the `READY`-state `sync` commands; lower-level `load/expand/-validate` stages are internal steps of those workflows.

Action	Minimum state	Produces
<code>initialise(.cgs)</code>	none	clone tree + READY
<code>clean-init(.cgs)</code>	none	purge generated state + clone tree + READY
<code>purge(.cgs)</code>	none	removed generated root-level clone state
<code>initialise(.gts)</code>	none	loaded/restored READY
<code>pull(.cgs .gts)</code>	READY	resynchronised READY
<code>checkout(.gts)</code>	READY	READY
<code>add</code>	READY only	READY
<code>commit <msg></code>	READY only	READY
<code>push</code>	READY only	READY + updated hash
<code>tag <name></code>	READY only	READY + updated tag
<code>freeze</code>	READY only	next <code>.gts</code> id

For `.cgs` refs, repositories can be declared by branch or by tag. Format validation is static and does not resolve either reference remotely. The runtime layer selects the declared target (tag takes precedence when both are present) and checks its availability through `GitRunner`.

The authoring boundary is explicitly `PARSE -> NORMALIZE -> VALIDATE`. TOML parsing produces authoring data; normalization expands `provider:owner/repository` identifiers and fills `main` branch defaults, `ssh`, automatic nested discovery, and relative paths. Validation receives only the complete canonical `CgsDocument`. Authoring syntax is therefore not the internal representation.

File and CLI authoring converge at this boundary. The CLI collects `-project` and repeatable `-repo` values, then calls the non-interactive `ComplexGitSyncClient.configure()` Python API. Python callers can use the same method directly. The facade delegates to `CgsDocument`; neither it nor the CLI owns repository grammar or normalization. Loading an equivalent `.cgs` file and using either definition path produce semantically identical canonical documents. `create-cgs` runs the same offline pipeline and serializes the result without entering Git runtime.

The boundary is also responsible for the reverse projection. `CgsDocument.to_git_tree()` creates the reference model and `CgsDocument.from_git_tree()` converts a reference or working tree back to canonical configuration. `GitTree.to_cgs()` remains only as a convenience delegate; it does not assemble repository tables or format TOML. The minimal serializer then removes deterministically reconstructible defaults. After the generated TOML is parsed and normalized again, its canonical representation must equal the one before serialization. Byte equality is not part of this contract.

Repository placement is parent-relative. At the top level the parent path is the project root; during nested discovery it is the repository whose `.cgs` is being expanded. Thus `../sibling` resolves to an existing sibling directory. Discovery compares normalized absolute paths and keeps an existing registry entry instead of inserting a duplicate.

Placement and traversal are independent. `nested_config = auto` searches the repository root for exactly one `*.cgs` — finding zero resolves the repository as a normal leaf, finding more than one raises immediately — `disabled` stops discovery through that reference, and a relative `.cgs` path selects an exact file inside the repository, failing if that named file does not exist. Explicit paths may not escape the repository root.

The textual `provider:owner/repository` grammar belongs exclusively to `cgs_format.py`. Its `parse_repo_id()` function owns syntax validation and splitting. Normalization uses that function, and CLI repository arguments must reuse it rather than introduce another parser.

The complete `.cgs` format pipeline is deterministic and offline. Parsing, normalization, validation, and serialization do not probe repositories or execute Git. Remote existence and branch/tag availability are resolved only after the validated document enters the runtime layer, through explicit `GitRunner` operations.

Provider behavior remains in `git_repo.py`: the `GitProvider` enum, canonical provider set, known-host map, and `RepoAddress` URL composer operate on already-separated identity fields. GitHub, GitLab, and Codeberg map respectively to `github.com`, `gitlab.com`, and `codeberg.org`. `custom` has no known-host entry and requires an explicit `gitprovider_url`; no fallback host is guessed.

Before any mutating workflow step (`commit`, `push`, `tag`, or `freeze`), the action layer now runs a workspace preflight validator. It checks dirty worktrees, detached HEADs, missing remotes, branch divergence, unresolved merges, and stale recorded `commit_sha` values. Diagnostics are severity-based: warnings report actionable-but-allowed states, while blocking errors stop the operation. `tag` still requires a clean worktree and a non-existing tag.

Formal `.gts` snapshot specification (T31)

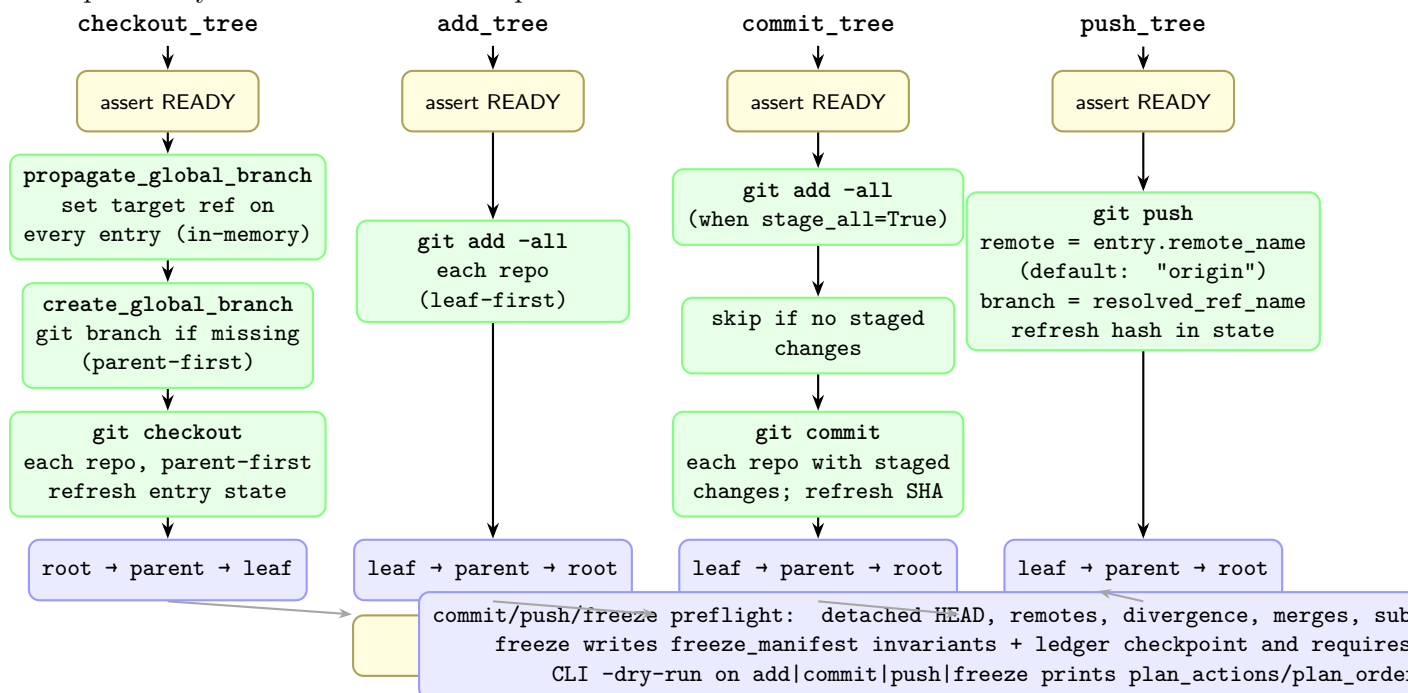
`.gts` now carries explicit schema and deterministic hash metadata in the `[document]` table: `schema_version = "1.1"`, `hash_algorithm = "sha256"`, and `snapshot_hash = "<64-hex>"`.

The canonical digest is computed from a deterministic payload made of `project`, `tree_state`, and sorted `repo_state` records, excluding volatile generation metadata (`generated_at` and `command_origin`). Non-root entries require `parent_absolute_path`; READY repositories require `commit_sha`. This makes `.gts` a stable checkpoint primitive for repeatable workspace state comparison.

Each `repo_state` record also carries the repository’s provider identity: `gitprovider`, `group_name`, and `gitprovider_url` (mirroring the same fields a `.cgs` repo entry carries — §5 “Local Authoring Spec”), plus `access_protocol`. `gitprovider/group_name/gitprovider_url` are part of the canonical hashed payload: they say *which repository this is*, so two snapshots pointing at different hosts are not the same state. `access_protocol` is deliberately excluded from the hash — it records only how the repository was last reached (ssh or https), not what it is, and a tree cloned entirely over ssh must hash identically to the same tree cloned over https. A `.gts` written before this schema addition has no `gitprovider` field at all; every command that would otherwise derive a clone URL from it (only the missing-repository restore path — `push/pull` instead convert the repository’s actual existing remote in place, never rebuilding it from identity) refuses with an explicit error instead of silently defaulting to GitHub.

Figure 3.4 shows the internal sequence of the four tree-wide synchronisation actions implemented in `operations.py`.

Figure 3.4: Synchronisation operations: `pull/checkout_tree` process `ROOT` → `PARENT` → `LEAF`; `add_tree`, `commit_tree`, and `push_tree` iterate `LEAF` → `PARENT` → `ROOT` so that inner repos are synchronized before their parents.



3.5 Tier 3 — Client / API

`ComplexGitSyncClient` is the single public facade. It:

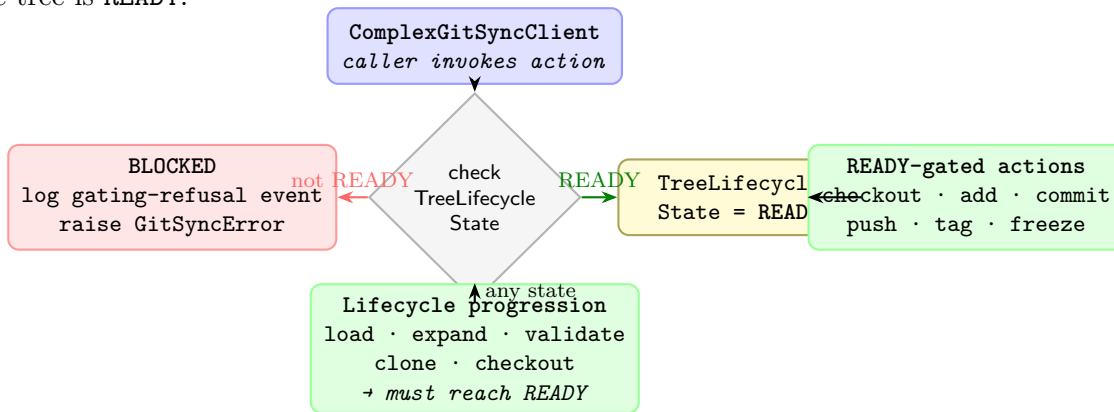
- holds references to `Orchestre`, `GitRunner`, `RuntimeStateStore`, and the live `WorkingGitTree`;
- computes the current `TreeLifecycleState` on demand and gates every action against it;

- emits structured log events for every state transition, action start/end, fallback decision, and .gts write/load.

Orchestre is the coordination layer between the Client and the tree models. The CLI (`cli.py`) collects terminal arguments or interactive prompt values, then delegates to the reusable Client API. The Client delegates format semantics to `cgs_format.py`; runtime operations remain separate.

Figure 3.5 shows how the Client gates action access based on the live tree state.

Figure 3.5: Client state-gating: the Client reads `TreeLifecycleState` before delegating to an action. **READY**-gated actions (`checkout`, `add`, `commit`, `push`, `tag`, `freeze`) are blocked unless the tree is **READY**.



3.6 Module Layout by Tier

Listing 3.1: Source modules grouped by architectural tier

```

src/ComplexGitSync/
# --- Tier 1: Core Data (centred on GitTree / GitRepo) ---
git_repo.py          GitRepo, WorkingRepo, RepoAddress,
                     RepoNode, enums
git_tree.py          GitTree, WorkingGitTree,
                     ProjectTreeState, tree utilities,
                     sync_gitignore (.gitignore maintenance)

# --- Tier 2: Actions ---
operations.py        propagate_global_branch,
                     checkout_tree, commit_tree, push_tree
create_global_branch
orchestre.py         discover_nested_configs(),
                     build_registry_from_cgs_document, render
                     helpers

# --- Tier 3: Client / API ---
orchestre.py         Orchestre, ComplexGitSyncClient
cli.py               build_parser / main
orchestre.py         GitRunner, RuntimeStateStore,
                     CommandRunLogger + create_run_logger
master.py            MasterConfig (workspace-local Git
                     identity
                     for ComplexGitSync-authored commits)
  
```

```
# --- Document layer (cross-cutting) ---
config_document.py          ConfigDocument (format-neutral base)
cgs_format.py               CgsDocument (.cgs parse/normalize/
                             validate/serialize)
orchestre.py                GtsDocument
errors.py                   exception hierarchy
```

3.7 Document Hierarchy

Figure 3.8 shows the document class hierarchy.

All persistent documents share the `ConfigDocument` base class, which provides format-agnostic serialisation (`to_toml`, `to_json`, `to_yaml`) and factory methods (`from_toml`, `from_json`, `from_yaml`). Subclasses add document-specific validation and convenience properties. The base is defined in `config_document.py`; `cgs_format.py` owns the complete `.cgs`-specific boundary, while `orchestre.py` consumes validated documents and retains runtime/orchestration responsibilities.

- **CgsDocument** (`.cgs`) — the authoring spec; describes the static project topology. *Never* a runtime snapshot.
- **GtsDocument** (`.gts`) — a generated snapshot capturing the exact state of the tree including commit SHAs and absolute paths. *Never* hand-edited.
- **Local Git Register** (`.lgr`) — the project-local register that assigns one stable local id to each generated `.gts` and tracks the current snapshot pointer.

3.8 Registry Model

Figure 3.9 shows the registry model.

`WorkingGitTree` is the authoritative in-memory graph. It maps *repo IDs* (colon-separated path strings such as `root:deps/child-repo`) to `WorkingRepo` instances. `WorkingRepo` holds both the static identity declared in `.cgs` and the dynamic state updated during operations.

Builder functions in `orchestre.py` translate document objects into a live working tree:

```
# Load from authoring spec
registry = build_registry_from_cgs_document(document, config_path)

# Load from snapshot
registry = build_registry_from_gts_document(snapshot_document)

# Serialise to snapshot
gts_doc = build_gts_document_from_registry(
    registry, command_origin="clone", source_cgs_path=path
)
```

Figure 3.6 is deliberately kept at Tier granularity rather than diagramming all of `src/ComplexGitSync`’s current modules — see `docs/DevGuide/architecture.md` for the dev-facing companion that carries the full per-module Ring-level breakdown and dependency graph this chapter does not attempt to reproduce.

Figure 3.6: Class overview by tier. Headline classes for each of the three tiers, plus the cross-cutting Document layer and the cli/ adapter package that the Tier model never described. Every module in `src/ComplexGitSync/` is named here, either as a class box or in a tier's module list; see `docs/DevGuide/architecture.md` for the full current module-by-module Ring-level dependency graph.

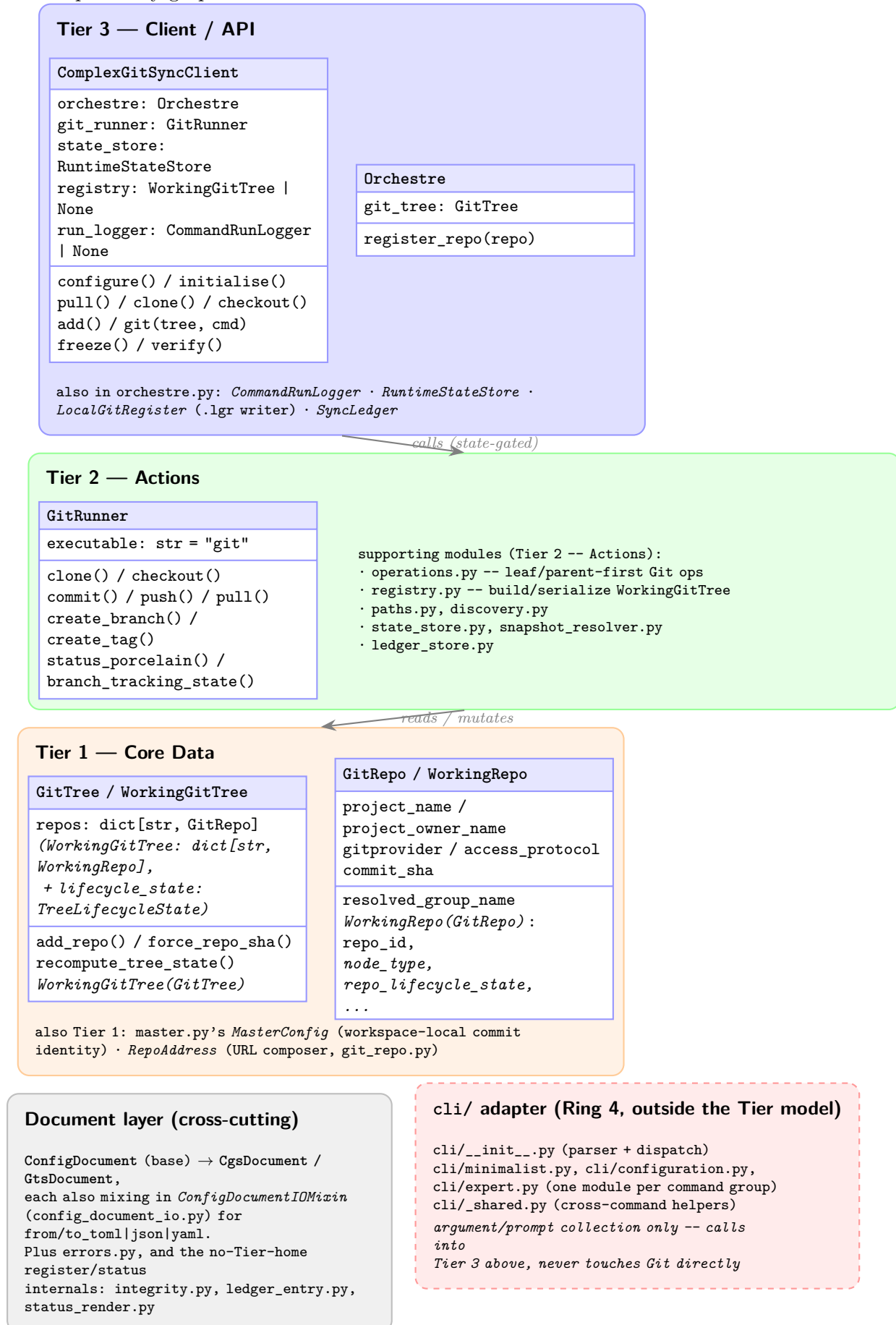


Figure 3.7: GitTree lifecycle state machine. The simplified user-facing lifecycle is `initialise(.cgs) → clone → READY`; `initialise(.gts) → restore → READY`; `pull(.cgs|.gts) → resync → READY`. Synchronized Git operations (`checkout`, `add`, `git("commit")`, `git("push")`, `git("tag")`, `freeze`) are gated on `READY`.

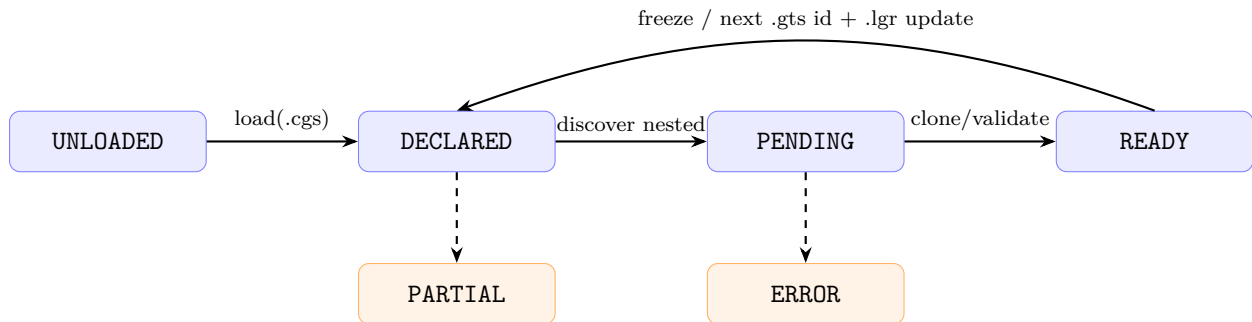


Figure 3.8: Document class hierarchy. `ConfigDocument` (`config_document.py`) is the abstract, I/O-free base; `ConfigDocumentIOMixin` (`config_document_io.py`) supplies the file-based `from/to_toml|json|yaml` methods. Each concrete document type combines both via ordinary multiple inheritance.

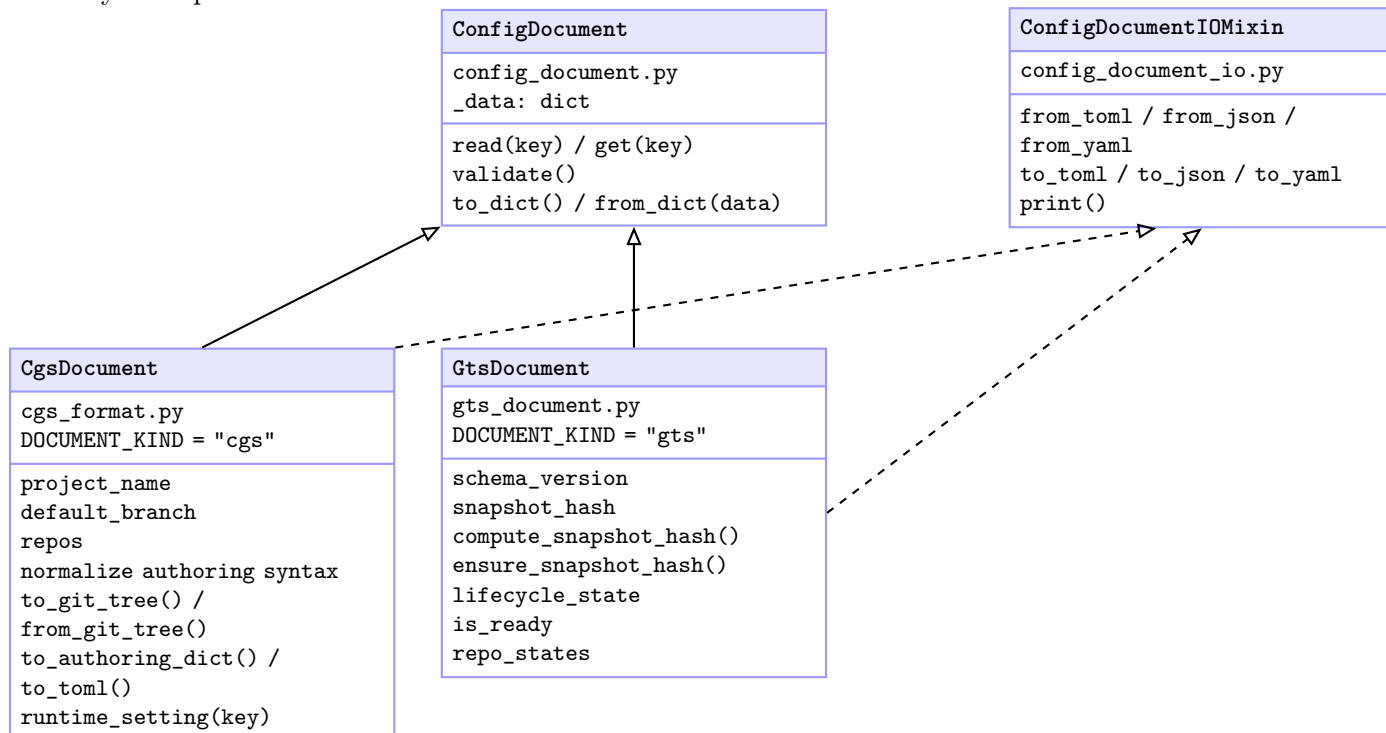
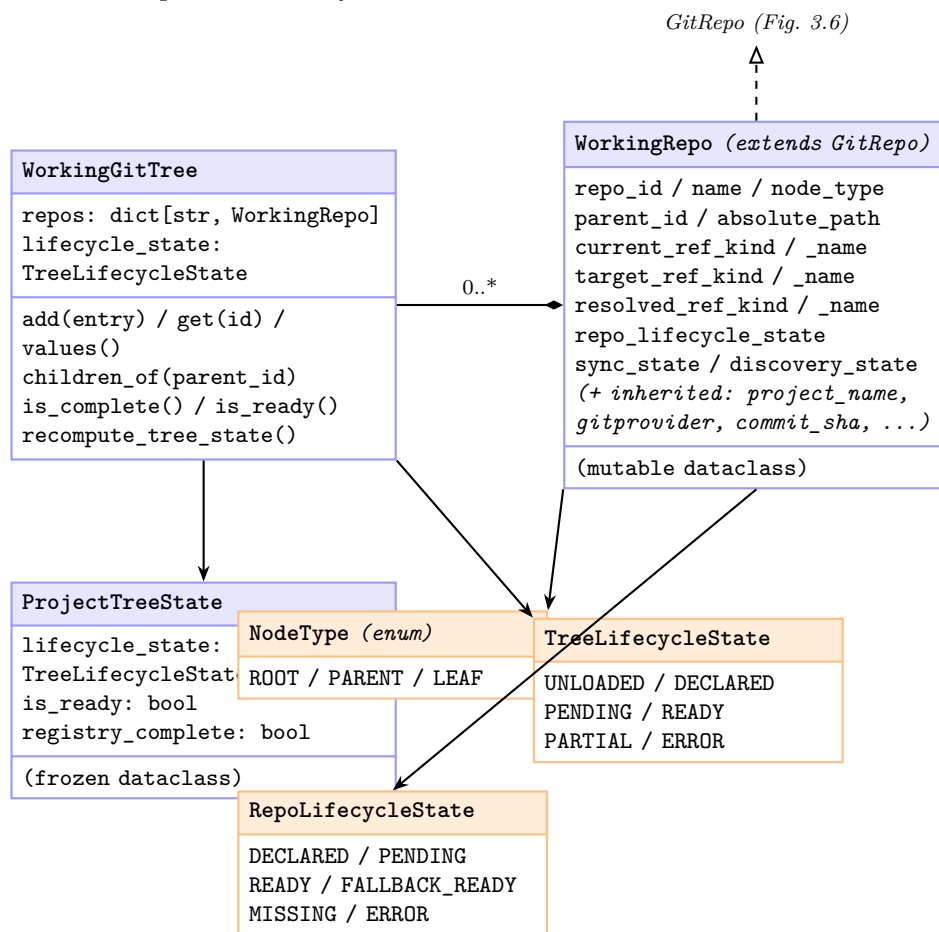


Figure 3.9: Registry model. `WorkingGitTree` is the authoritative in-memory graph; `WorkingRepo` *extends* `GitRepo` (Figure 3.6) rather than duplicating its identity fields, adding only the mutable tree-position and synchronisation state below.



Chapter 4

Writing `cgs` Files: The Complete Authoring Guide

This section is the definitive reference for authoring `.cgs` (ComplexGitSync specification) files. It covers both the simplified shorthand syntax suitable for most use cases and the full advanced syntax for complex topologies, with practical guidelines, complete examples, and common pitfalls.

4.0.1 What is a `cgs` File?

A `.cgs` file is a **TOML-formatted** specification that describes a tree of Git repositories to be managed as a single logical unit. It is the only hand-authored input in ComplexGitSync; all other documents (`.gts`, `.lgr`) are generated automatically at runtime.

Key properties:

- **Offline and deterministic:** Parsing, normalization, and validation require no network access. The same file always produces the same internal representation.
- **Never modified at runtime:** `.cgs` files are read-only during execution. Snapshots (`.gts`) capture workspace state; they do not update the specification.
- **Two equivalent forms:** A concise shorthand syntax for simple cases, and a verbose inline-table syntax for advanced overrides. Both normalize to the same canonical internal document.
- **Validation is strict:** Unknown fields, malformed identifiers, duplicates, and missing required keys are rejected before any Git operations begin.

4.0.2 Minimal `cgs`: The Simplified Form

For the vast majority of projects, a `.cgs` file needs only two top-level keys: `project` (a string) and `repos` (a list of repository identifiers).

Basic Structure

```
project = "my-project"

repos = [
  "github:owner/repo1",
  "gitlab:group/repo2",
  "codeberg:user/repo3",
]
```

Repository Identifier Syntax

Each repository identifier follows the pattern:

`provider:owner/repository`

provider One of the supported Git providers: `github`, `gitlab`, `codeberg`, or `custom`.

owner The owner (user or organization/group) on the provider.

repository The repository name itself.

Nested namespace paths (e.g., GitLab subgroups) are supported. The final path segment is the repository name; all earlier segments join into `project_owner_name`:

```
repos = [
  "gitlab:my-group/my-subgroup/my-repo", % 3-level namespace
  "github:my-org/my-repo",               % 2-level namespace
]
```

Valid providers and their default hosts:

Provider	Host	Remotes Generated
github	github.com	SSH and HTTPS
gitlab	gitlab.com	SSH and HTTPS
codeberg	codeberg.org	SSH and HTTPS
custom	explicit URL required	SSH and HTTPS

4.0.3 Normalization: What the Simplified Form Expands To

The simplified shorthand is automatically **normalized** into a complete canonical document. Every omitted field receives a deterministic default value.

Default Values Applied During Normalization

Field	Default	Applied To
<code>format_version</code>	"1.0"	document table
<code>default_branch</code>	"main"	project and each repo
<code>fallback_branch</code>	"main"	each repo
<code>access_protocol</code>	"ssh"	each repo
<code>nested_config</code>	"auto"	each repo
<code>relative_path</code>	repo name	each repo

Normalization Example

The following simplified input:

```
project = "demo"

repos = [
  "github:acme/core",
  "gitlab:acme/utils",
]
```

Normalizes to this canonical form (conceptually):

```
[document]
format_version = "1.0"

[project]
name = "demo"
default_branch = "main"

[[repos]]
gitprovider = "github"
project_owner_name = "acme"
project_name = "core"
repo_name = "core"
default_branch = "main"
fallback_branch = "main"
access_protocol = "ssh"
nested_config = "auto"
relative_path = "core"

[[repos]]
gitprovider = "gitlab"
project_owner_name = "acme"
project_name = "utils"
repo_name = "utils"
default_branch = "main"
fallback_branch = "main"
access_protocol = "ssh"
nested_config = "auto"
relative_path = "utils"
```

Important: The project's `default_branch` propagates to all repos unless explicitly overridden per-repo.

4.0.4 Inline Table Syntax: Overriding Defaults

When a repository needs non-default values for any field, use an **inline table** instead of a plain string. Only the fields that differ from defaults need to be specified.

Common Override Scenarios

```
project = "demo"

repos = [
    % Non-default branch
    { repository = "github:acme/core", default_branch = "develop" },

    % Different default and fallback branches
    { repository = "gitlab:acme/libs",
      default_branch = "next",
      fallback_branch = "main" },

    % Custom placement path
    { repository = "github:acme/docs",
      relative_path = "documentation" },

    % HTTPS access instead of SSH
    { repository = "codeberg:org/tool",
      access_protocol = "https" },
```

```

% Disable nested config discovery
{ repository = "github:acme/legacy",
  nested_config = "disabled" },

% Custom provider with explicit URL
{ repository = "custom:team/internal",
  gitprovider_url = "https://git.internal.company.com" },
]

```

Combined Overrides

Multiple overrides can be combined in a single inline table:

```

repos = [
  { repository = "github:acme/web",
    relative_path = "apps/web",
    default_branch = "production",
    fallback_branch = "staging",
    access_protocol = "https",
    nested_config = "disabled" },
]

```

4.0.5 Advanced Table Syntax: Legacy Form

The legacy advanced form uses explicit TOML tables. Fully supported but not recommended for new files.

```

[project]
name = "demo"
default_branch = "main"

[[repos]]
repository = "github:acme/core"

[[repos]]
repository = "gitlab:acme/utils"
default_branch = "develop"
relative_path = "lib/utils"

```

When to use advanced tables:

- Project-wide settings in the [project] table
- When you prefer visual separation between entries
- For compatibility with old ComplexGitSync versions

Note: The [project] table can also be inline: `project = { name = "demo", default_branch = "main" }`.

4.0.6 Repository Placement and Path Resolution

relative_path: Where Each Repository Lives

The `relative_path` field determines where a repository is cloned relative to its parent. Default: the repository name.

```
% Default: repo cloned at <parent>/repo_name
repos = ["github:acme/core"]

% Explicit: repo cloned at <parent>/apps/core
repos = [
    { repository = "github:acme/core", relative_path = "apps/core" }
]
```

Project Root Auto-Mount Rule

If exactly one repository name matches the project name, its `relative_path` is automatically set to `"."`:

```
project = "CGSil1"

repos = [
    "gitlab:CGS_test/CGSil1", % Auto-mounted at "."
    "gitlab:CGS_test/CGSil2", % At "./CGSil2"
    "github:flipoyo/CGSih1",  % At "./CGSih1"
]
```

Equivalent to:

```
project = "CGSil1"
repos = [
    { repository = "gitlab:CGS_test/CGSil1", relative_path = "." },
    { repository = "gitlab:CGS_test/CGSil2", relative_path = "CGSil2" },
    { repository = "github:flipoyo/CGSih1", relative_path = "CGSih1" },
]
```

Cross-Repository References

A repository can reference another repository in the same tree using relative paths:

```
% In CGSil2.cgs (nested under CGSil2/):
repos = [
    { repository = "github:flipoyo/CGSih1",
      relative_path = "../CGSih1" }
]
```

Here, `../CGSih1` resolves from `<CGSHOME>/CGSil2/` to `<CGSHOME>/CGSih1`, referencing the sibling declared elsewhere. No `nested_config` override is needed: discovery's absolute-path dedup guard already skips re-registering `CGSih1` once it is known under its canonical entry, before `"auto"` ever gets a chance to reopen its `.cgs`.

4.0.7 Nested Configuration Discovery

The `nested_config` field controls whether `ComplexGitSync` looks for additional `.cgs` files inside each repository.

auto (default) Loads the sole root-level `*.cgs` file if one exists; finding none is a normal leaf, not an error; multiple matches are rejected.

disabled Does not inspect the repository for nested config.

a relative path A path ending in `.cgs` loads that exact file; if the file doesn't exist, this is treated as an error, since an explicit path asserts that a specific file must be there.

Example: Multi-Level Nested Topology

```
project = "parent-project"

repos = [
    "github:org/parent",
    { repository = "github:org/child-a", nested_config = "auto" },
    { repository = "github:org/child-b", nested_config = "disabled" },
    { repository = "github:org/child-c", nested_config = "config/prod.
      cgs" },
]
```

4.0.8 Access Protocols and Custom Providers**SSH vs HTTPS**

Default: SSH. To use HTTPS:

```
repos = [
    { repository = "github:owner/repo", access_protocol = "https" },
]
```

Generated remotes:

- SSH: `git@github.com:owner/repo.git`
- HTTPS: `https://github.com/owner/repo.git`

Custom Git Providers

For providers not in the built-in list, use custom with explicit URL:

```
repos = [
    { repository = "custom:team/project",
      gitprovider_url = "https://git.internal.mycompany.com" },
]
```

Important: `gitprovider_url` is **required** for custom.

```
% INCORRECT - will fail validation
repos = ["custom:team/project"]

% CORRECT
repos = [
    { repository = "custom:team/project",
      gitprovider_url = "https://git.internal.mycompany.com" }
]
```

4.0.9 Branches: default_branch and fallback_branch**Branch Resolution Order**

When cloning or pulling:

1. Repository's `default_branch`
2. Project's `default_branch`
3. Repository's `fallback_branch`
4. Project's `fallback_branch` (defaults to `main`)

Practical Branch Patterns

```
% All repos use "main" (default)
project = "my-project"
repos = ["github:owner/repo1", "github:owner/repo2"]

% All repos default to "develop"
project = { name = "my-project", default_branch = "develop" }
repos = ["github:owner/repo1", "github:owner/repo2"]

% Per-repo overrides
project = "my-project"
repos = [
  "github:owner/core", % Uses default "main"
  { repository = "github:owner/feature", default_branch = "dev" },
  { repository = "github:owner/stable",
    default_branch = "release", fallback_branch = "main" },
]
```

4.0.10 Private Repositories

A tree holds two kinds of repository. **Project repos** are the work itself and follow the project's branch. **Private repos** are how the project is run — pipelines, agent instructions, house rules — and are shared with the author's other projects, so they must not follow it: creating a project's feature branch inside a repository other projects also mount pollutes them all.

Setting `private = true` keeps a repository off the project's branch. Adding `writable = true` makes it **private/local** — this project's own settings, on a branch named after the project, which `cgitsync` may write to. Without it the repository is **private/distant**: read-only, because it belongs to someone else.

```
repos = [
  "github:flipoyo/ComplexGitSync",
  { repository = "github:flipoyo/.agentSpec", default_branch = "main",
    nested_config = "auto", private = true },
  { repository = "github:flipoyo/.localSpec", default_branch = "
    MyProject",
    fallback_branch = "main", private = true, writable = true },
]
```

With that in place, `cgitsync branch feature-x` creates `feature-x` in the project's own repositories and in neither mount; `cgitsync checkout` leaves both on their declared branch; and `cgitsync pull` still updates them, on that branch rather than the tree's.

Two limits are worth knowing. Pinning governs *branch* propagation only: restoring a frozen release with `launch-release` puts every repository back on the commit the snapshot records, private or not, so a release stays reproducible. And a repository declared in a *nested* `cgis` is private by that file, which lives in the repository that owns it — pinning it means publishing that change there, not in the parent project.

4.0.11 Complete Reference: All Recognized Fields

Top-Level Fields

Field	Type	Required	Default
<code>project</code>	string or table	yes	—
<code>repos</code>	list	yes	—
<code>document</code>	table	no	<code>{format_version = "1.0"}</code>

Project Table Fields

Field	Type	Required	Default
name	string	yes	—
default_branch	string	no	"main"

Repository Fields

Field	Type	Required	Default
repository	string	yes*	—
gitprovider	string	no	from repository
project_owner_name	string	no	from repository
project_name	string	no	from repository
repo_name	string	no	from repository
default_branch	string	no	from project or "main"
fallback_branch	string	no	"main"
access_protocol	string	no	"ssh"
relative_path	string	no	repo name (or "." if matches project)
nested_config	string	no	"auto"
private	boolean	no	false
gitprovider_url	string	no**	—

* Either `repository` (shorthand) or the combination of `gitprovider`, `project_owner_name`, `project_name` is required.

** `gitprovider_url` is required when `gitprovider` = "custom".

4.0.12 Practical Examples**Example 1: Simple Monorepo-Style Project**

```
project = "my-app"

repos = [
    "github:myorg/my-app",    % Root (auto-mounted at ".")
    "github:myorg/core",     % At ./core
    "github:myorg/ui",       % At ./ui
    "github:myorg/api",      % At ./api
]
```

Example 2: Mixed Providers with Custom Branches

```
project = "cross-provider"

repos = [
    "github:myorg/main-repo",
    { repository = "gitlab:mygroup/data", default_branch = "data-main" },
    { repository = "codeberg:user/tools", access_protocol = "https" },
]
```

Example 3: Complex Topology with Nested Configs

```
project = "big-system"

repos = [
  "github:org/root",
  { repository = "github:org/frontend",
    relative_path = "apps/frontend", nested_config = "auto" },
  { repository = "github:org/database",
    relative_path = "infra/db", nested_config = "disabled" },
  { repository = "github:thirdparty/lib",
    relative_path = "external/thirdparty-lib", nested_config = "
      disabled" },
]
```

Example 4: Real-World (cawaqsviz)

Based on `examples/cawaqsviz.cgs`:

```
project = { name = "CaWaQS-Viz", default_branch = "main" }

repos = [
  { repository = "gitlab:cawaqs/gviz/cawaqsviz",
    relative_path = ".", fallback_branch = "main" },
  { repository = "github:flipoyo/HydrologicalTwinAlphaSeries",
    relative_path = "external/HydrologicalTwinAlphaSeries",
    fallback_branch = "main" },
  { repository = "github:flipoyo/hydrological_twin",
    relative_path = "external/HydrologicalTwinAlphaSeries/docs/
      hydrological_twin",
    fallback_branch = "main" },
  { repository = "github:flipoyo/user_guide_CaWaQS-Viz",
    relative_path = "docs/CWV_user_guide",
    fallback_branch = "main" },
]
```

Four repositories on two levels. The third entry's path lies inside the second's, which is how a `.cgs` says that one repository holds another: every path is written from the project root, so nesting shows up as one path lying inside another, and the innermost holder wins. `HydrologicalTwinAlphaSeries` is therefore read back as a `PARENT` with `hydrological_twin` as its child.

No repository here has a `.cgs` of its own, but no `nested_config` override is written for that: the default `"auto"` already resolves a repository with zero nested `*.cgs` matches cleanly.

4.0.13 Authoring with CLI Tools

You do not need to write `.cgs` files by hand.

discover: Scan an Existing Checkout

```
$ pixi run cgitssync discover ~/work/myproject --write draft.cgs
$ pixi run cgitssync validate draft.cgs
```

configure: Interactive Prompt

```
$ pixi run cgitssync configure --output myproject.cgs
```

create-cgs: Direct CLI Authoring

```
$ pixi run cgitssync create-cgs \
  --project myproject \
  --repo github:owner/repo1 \
  --repo gitlab:group/repo2 \
  --output myproject.cgs
```

All three use the same offline pipeline as direct TOML parsing.

4.0.14 Validation and Common Errors

Always validate before use:

```
$ pixi run cgitssync validate myproject.cgs
```

Validation Checks

- Presence of required keys (project, repos)
- Valid repository identifier syntax
- Known provider names or custom with URL
- custom providers have gitprovider_url
- Valid nested_config values
- Valid access_protocol values
- No duplicate repository declarations
- Non-empty strings for all text fields

Common Errors and Fixes

project is required Add project = "name".

repos is required Add repos = [...].

expected 'provider:owner/repository' Fix identifier syntax.

gitprovider_url is required for custom provider Add gitprovider_url = "https://...".

duplicate repository Remove one or make relative_path distinct.

nested_config must be auto, disabled, or a path Use valid value. Paths must end in .cgs.

4.0.15 Templates and Starting Points**Template Files**

examples/template.cgs Copy for normal authoring.

examples/normalized_template.cgs Reference for canonical form.

Quick Start Checklist

1. Copy `examples/template.cgs` to `<your-project>.cgs`
2. Edit `project` name
3. Add repositories to `repos` list
4. Override defaults only when necessary
5. Validate: `pixi run cgitssync validate <file>.cgs`
6. Initialize: `pixi run cgitssync initialise <file>.cgs`

4.0.16 Best Practices

- **Start simple:** Use plain strings until you need overrides.
- **Use auto-mount:** Name one repo after project for automatic root.
- **Validate early and often:** After every edit.
- **Keep comments:** Document non-default values.
- **Avoid advanced tables:** Inline tables are more maintainable.
- **Use `nested_config = "disabled"` judiciously:** The default `"auto"` already resolves cleanly whether or not a repo has its own `.cgs` — including cross-references and cycle back-references, which the discovery dedup guard already prevents from re-registering. Reserve `"disabled"` for the rarer case where a repo does carry a `.cgs` you explicitly want `ComplexGitSync` to never open.
- **Prefer SSH:** Works well for most providers.
- **Be explicit with custom:** Always include `gitprovider_url`.

4.0.17 Internal Round-Trip Guarantee

`ComplexGitSync` guarantees a semantic round-trip:

$$.cgs \rightarrow \text{CgsDocument} \rightarrow \text{GitTree} \rightarrow \text{CgsDocument} \rightarrow .cgs$$

Parsing and normalizing a generated `.cgs` reproduces the canonical document. Whitespace and comments need not match byte-for-byte, but semantics must.

Chapter 5

User Guide

This chapter is the authoritative reference for every user-facing command, configuration option, document format, and error condition exposed by **ComplexGitSync**.

5.1 CLI Overview

The single entry-point is the `cgitsync` command.

cgitsync is a Pixi task, not a globally installed executable. This project is developed and run exclusively through Pixi (`pixi install` once, per checkout) — there is no `pip install` that puts a bare `cgitsync` on `$PATH`. Every invocation shown in this chapter as `pixi run cgitsync ...` must be typed exactly that way; a bare `cgitsync ...` will not be found by the shell.

```
$ pixi run cgitsync <command> [options]
```

Running `cgitsync` without a command prints a help summary and exits with code 0.

5.1.1 Prerequisites

Three tools are needed before the first command: Git clones this checkout and the repositories in a tree; Pixi installs the locked Python environment and runs `cgitsync`; and repository creation needs the provider’s own command line tool — `gh` for GitHub, `glab` for GitLab, or `tea` for Codeberg. Install only the provider tool you use.

The complete list depends on the tree and the work being done:

Dependency	When it is needed	Where its version or requirement comes from
Git	Every clone, inspection and synchronisation.	Its observed version is recorded with the State.
Pixi	Install this project's environment and run <code>cgitsync</code> .	<code>pixi.lock</code> pins the project environment; the observed Pixi version is recorded separately.
Python	Run <code>ComplexGitSync</code> inside the Pixi environment.	<code>pixi.toml</code> selects the interpreter; the observed interpreter is recorded.
<code>gh</code> , <code>glab</code> , or <code>tea</code>	<code>repo create</code> , according to the repository provider.	The provider tool reports its version and whether it is authenticated.
SSH client and agent	Repositories whose remote uses SSH.	Presence is observed. Keys, user names, and key paths are never recorded.
DVC or Git LFS	A tree that uses the corresponding data backend.	The tree declares its use; the observed tool version is recorded when used.
Compilers and system libraries	Native builds or software installed outside Pixi.	The tree declares these in its <code>.cgs [environment]</code> table.
<code>latexmk</code> and <code>TeX</code>	Developers rebuilding the manuals under <code>docs/</code> .	A developer requirement, not an end-user runtime dependency.

The machine must also match a platform supported by the tree's lock files. `pixi run cgitsync env` reports the operating system, architecture, Pixi platform, libc, tool versions, credential facts, and manifests it observes. `pixi run cgitsync env check` compares that observation with the tree's declared requirements. Neither command installs anything.

`ComplexGitSync` stores no credentials of its own and has no private-repository authentication story: every git operation — `clone`, `pull`, `push`, and the remote lookups behind `checkout`'s branch/tag fallback — runs against whatever the ambient environment already has cached (`ssh-agent`, an HTTPS credential helper, a stored personal-access-token). When that ambient credential is missing or stale, the operation fails immediately with a normal, catchable error rather than sitting silently on a terminal or GUI credential prompt that will never arrive — every git subprocess `ComplexGitSync` runs sets `GIT_TERMINAL_PROMPT=0` and a no-op `GIT_ASKPASS` for exactly this reason. Fix the ambient credential (switch the remote to `ssh`, refresh the token) and re-run the command; nothing needs to be undone first.

5.2 Lifecycle Contract

This guide follows the simplified canonical lifecycle:

1. `initialise(.cgs)` → clone all repos → **READY** (*new project*)
`initialise(.gts)` → restore from snapshot → **READY** (*existing project*)
2. `pull(.cgs|.gts)` → resync an existing tree
3. `checkout(.gts)` / `add` / `commit` / `push` / `tag` → git operations on the tree
4. `freeze` → emit the next `.gts` id and update `<Project_name>.lgr`

The project-local `.lgr` register and sync ledger are both active (T24, T32). Every synchronisation operation is automatically recorded as an immutable DAG event in the `[[ledger]]` section of the `.lgr` file.

5.3 Commands

The CLI exposes three command groups once a `GitTree` is `READY`. The minimalist level is `initialise`, `bootstrap`, `clean-init`, `freeze-release`, `freeze-release-force`, `status`, `view-tree`, and `launch-release`. The expert level exposes the individual operations: `branch`, `checkout`, `purge`, `validate`, `clone`, `pull`, `pull-force`, `add`, `commit`, `push`, `tag`, `freeze`, `import-submodules`, `init-from-submodules`, and `verify`, plus `env` and `env check`. The configuration level, `discover`, `configure` and `create-cgs`, authors a `.cgs` specification without requiring existing state. Redundant inspection aliases such as `print`, `view-operation`, and `validate-topology` are not part of the public CLI surface.

5.3.1 initialise

```
$ pixi run cgitssync initialise <spec.cgs> [--output-path DIR] [--commit-
  gitignore]
  [--force-gitignore-sync] [--git-user-name NAME] [--git-user-email
    EMAIL]
  [--force-protocol {ssh,https}] [--force-reclone]
$ pixi run cgitssync initialise <snapshot.gts>
$ pixi run cgitssync initialise --project NAME --repo PROVIDER:OWNER/REPO
  [--repo ...] [--output-path DIR]
```

Unified initialisation entry point. Dispatches on file extension:

- `.cgs` source (clone mode): clones the full repository tree. The optional `-output-path` controls `CGSPATH`; `CGSHOME` is derived as `CGSPATH/<project-name>`. When omitted, `CGSPATH` defaults to `../...`. On success, ends in `READY`. If clone setup fails because stale generated state is still present, the CLI prints `Try clean-init method`.
- `.gts` source (restore mode): loads a saved snapshot directly. On success, ends in the lifecycle state recorded in the snapshot.
- Direct CLI authoring: requires `-project` and at least one repeatable `-repo`. It is mutually exclusive with a source file. The CLI collects values only and calls `ComplexGitSyncClient.configure()`; that reusable Python API delegates parsing, normalization, and validation to `cgs_format.py` and produces the same canonical `CgsDocument` as an equivalent `.cgs` file.

All initialization paths end in a `READY` tree or fail explicitly.

initialise re-clones your dependencies. Only the root repository at `CGSHOME` is kept as it is. Every repository below it whose directory already holds files is **deleted and cloned again**. The old `.git` goes with it, so nothing in that directory can be recovered once the run has passed that point.

Before deleting anything, `initialise` inspects each destination and stops the whole run when one of them holds work that exists nowhere else: uncommitted changes, commits that have not been pushed to the branch's upstream, or a branch with no upstream at all. It names every repository that blocked it, in one message, and deletes none of them. The two ways forward are to commit and push that work, or to pass `-force-reclone` and delete it deliberately.

A destination that is not a Git checkout at all — what a clone interrupted halfway leaves behind — is still cleared without any flag. That is the case the deletion was written for, and it still works.

Before a `.cgs` source is confirmed ready, every repository with children (root, or any nested repository that itself has further nested children) is safely pulled parent-first and has its `.gitignore` updated with the relative path of each immediate child — nested repositories are plain independent clones rather than gitlinks, so without this a parent’s `git status/git add` would otherwise see a child’s working tree as ordinary untracked content. This is logged as the `GT-GITIGNORE` phase, after `GT-CLONE`. If the safe pull for one of these repositories fails, `initialise` raises immediately and nothing is written; pass `-force-gitignore-sync` to instead fall back to a pull-force recovery for that one repository (never a force-push). By default nothing is staged, committed, or pushed by this step — it only writes the file and prints what changed. Pass `-commit-gitignore` to explicitly approve staging (`.gitignore` alone), committing (with a message listing exactly which children were added), and pushing each changed repository. The commit identity defaults to local git config; `-git-user-name/-git-user-email` override it and persist the override to `CGSHOME/.cgitsync/master.toml` via `MasterConfig`, so later invocations on the same workspace pick it up without repeating the flags.

The root’s own `.gitignore` additionally always gains `.cgitsync/` (the generated runtime-state directory — snapshots, register, run logs, never part of the `.cgs/.gts` project spec) and its `<project-name>.lgr` register file, whether or not the tree has any nested repositories at all — that state is written under the root regardless of tree shape, so it is excluded the same idempotent way, via the same underlying write `sync_gitignore()` already uses for child mount points.

`-force-protocol {ssh,https}` is an expert option, typically used systematically in CI rather than for everyday development. It overrides `access_protocol` in memory for every repository this run clones — including ones discovered later from a nested `.cgs` living in a different, separately-cloned repository — so no `.cgs` file is ever read differently or written to. Forcing `https` is the usual CI case: it removes the dependency on an SSH key or agent being present on the runner, a dependency that otherwise differs between a `push` run and a `pull_request` run (GitHub Actions withholds repository secrets from a pull request opened from a fork). Omitted (the default), each repository follows whatever protocol its own `.cgs` entry declares, exactly as without the flag.

5.3.2 bootstrap

```
$ pixi run cgitsync bootstrap <spec.cgs> <project_name> [--cgs-path DIR]
  [--force-protocol {ssh,https}]
```

Clones the full repository tree from a `.cgs` source into an isolated `CGSHOME`, for running `cgitsync` from its own standalone clone rather than nested inside the project tree it manages — one `ComplexGitSync` clone reused across many projects. It delegates to the same `ComplexGitSyncClient.clone_cgs` pipeline as `clone` below, and so shares its branch-fallback and nested-discovery behaviour, but differs from both `clone` and `initialise` in two ways:

- `project_name` is a required positional argument, not inferred from the `.cgs` document. It always forms the final path segment of `CGSHOME`, regardless of the specification’s own project name.
- `CGSPATH` does not default to `../..` relative to the current directory (that default assumes `ComplexGitSync` is cloned inside the tree it manages — the nested-mode case `initialise` is for). When `-cgs-path` is omitted, `CGSPATH` instead defaults to a fresh `$HOME/.cgs/CGS<timestamp>/` directory, created automatically if it does not exist, so a bootstrapped project can never land inside `ComplexGitSync`’s own clone.

`-force-protocol {ssh,https}` behaves exactly as described under `initialise` above, applying here to every repository the run clones — including the root, which `bootstrap` (unlike `initialise`) also clones from scratch rather than assuming already checked out.

5.3.3 discover

```
$ pixi run cgitSync discover [ROOT] [--write FILE] [--max-depth N]
```

Scans `ROOT` (default: the current directory) for git repositories and drafts a `.cgs` from what is checked out, through the `ComplexGitSyncClient.discover_repos()` facade. This is the entry point for adopting a project that exists on disk but has no `.cgs` describing it yet.

The command is a pure read of the filesystem and of each repository’s git config: nothing is cloned, fetched, staged, or modified, and no network call is made. Without `-write` it only prints what it found, matching the “report first, write only when asked” posture of `-commit-gitignore` and `import-submodules -apply`.

The walk descends with no depth limit by default, treating every directory that contains a `.git` entry as a repository. Pass `-max-depth N` (`ROOT` itself is depth 0) to bound it. It never descends *into* a `.git` directory: for a submodule the real git directory lives at `<parent>/<.git/modules/<name>` while the child’s own `.git` is a file, so walking into it would report the same repository twice.

For each repository, `origin`’s URL is read and converted to the canonical `provider:owner/repository` shorthand through the existing `parse_repo_id()` grammar — this command adds no second parser. `relative_path` is taken straight from the walk rather than from a repository name, so a child mounted at `external/Thing` is recorded there and not at `Thing`. `nested_config` is left unset on every entry: the default `auto` already resolves cleanly whether or not a repository carries its own `*.cgs`.

Repositories with no `origin`, or whose remote URL does not resolve to a registered provider, are reported as warnings and left out of the draft. They are never guessed at: a draft that silently invented an address would be worse than one that states what it could not determine. Always review the generated file, then `validate` it.

A repository found *inside* another repository is drafted as that repository’s child, not the project root’s. Nesting is worked out by comparing paths: every path in a `.cgs` is written from the project root, so a repository whose path lies under another’s sits inside it, and the innermost one wins. The report marks such an entry `inside: <path>` and prints the resulting tree, and `build_registry_from_cgs_document()` reads the drafted file back the same way — the holding repository becomes a `PARENT` and keeps its own `.gitignore` entry for the repository it holds, which is what stops a later `add` from restaging that repository as a submodule.

Only what is *checked out at scan time* can be found. A repository cloned without `-recurse-submodules` leaves its submodule paths as empty directories, and those are correctly not reported; recovering them from git metadata instead is `import-submodules`’ job. The same applies when `-max-depth` is given: if it stops the walk while directories are still left to look into, the report says so in a warning rather than presenting a partial answer as a complete one. Without the flag, the walk cannot stop early — there is nothing to stop it at.

5.3.4 configure

```
$ pixi run cgitSync configure [--output FILE]
```

Interactively prompts for a project name, default branch, and one or more repositories (provider, owner, name, and per-repository overrides), then writes the resulting `.cgs` file through the same offline `ComplexGitSyncClient.configure()` facade used by `initialise -project/-repo` and `create-cgs`. The provider prompt accepts `github`, `gitlab`, `codeberg`, or `custom`; `custom` additionally prompts for the provider URL. When `-output` is omitted, the CLI prompts for a destination path, defaulting to `<project-name>.cgs`.

5.3.5 create-cgs

```
$ pixi run cgitSync create-cgs --project CGSil1 \
  --repo github:flipoyo/ComplexGitSync \
  --repo codeberg:GX4G/GX4G \
  --output CGSil1.cgs
```

Builds the canonical `CgsDocument` through the offline format pipeline and serializes concise `.cgs` TOML. It performs no Git or network work.

5.3.6 clean-init

```
$ pixi run cgitSync clean-init <spec.cgs> [--output-path DIR] [--commit-
  gitignore]
  [--force-gitignore-sync] [--git-user-name NAME] [--git-user-email
    EMAIL]
  [--force-protocol {ssh,https}]
```

Recovery-oriented initialisation for a `.cgs` source. It follows the same high-level pipeline as `initialise(.cgs)` (including the `.gitignore` lifecycle sync and `-force-protocol` behaviour described there), but inserts a cleanup phase before cloning:

load->expand->validate->purge->clone->gitignore

Use it when an earlier initialisation attempt left partial child checkouts. `CGSHOME/.cgitsync/master.toml` (the persisted Git identity override, if any) is preserved by the purge phase — it is workspace-local configuration, not generated clone state.

5.3.7 purge

```
$ pixi run cgitSync purge <spec.cgs> [--output-path DIR]
```

Removes generated clone state from the resolved `CGSHOME` without cloning. It deletes child repository directories declared directly under the project root and project-local `*.lgr` files. Nested directories that are not immediate root children, and `.cgitsync/master.toml` (see `clean-init` above), are left untouched by this command.

5.3.8 pull

```
$ pixi run cgitSync pull [spec.cgs|snapshot.gts] [--commit-gitignore]
  [--force-gitignore-sync] [--git-user-name NAME] [--git-user-email
    EMAIL]
  [--force-protocol {ssh,https}]
```

Resynchronises the project tree. `.cgs` sources reload the source, discover nested configs, then resynchronise the loaded tree. `.gts` sources load the saved registry as the starting point and then run the same pull workflow. In both cases the operation starts at the project root: `root -> parent -> leaf`. Every repository — root, parent, and leaf alike — is a plain independent clone and receives its own `git pull -ff-only`. Use `launch-release` when you want to check out a frozen recorded release state instead of pulling current remotes. If local changes block this safe pull, the CLI prints `You can try cgitSync pull-force command`.

Before pulling your own branch, `pull` fetches *every* branch of each remote. That is what lets `checkout` join a branch somebody else pushed rather than starting a new one of the same name (see `checkout` below). It also repairs a workspace whose clone left a single-branch fetch refspec behind, described under `clone`.

For a `.cgs` source, `pull` also runs the same `.gitignore` lifecycle sync described under `initialise` — including `-commit-gitignore`, `-force-gitignore-sync`, and the `-git-user-name/-git-user` identity override — once the tree-wide pull completes. A `.gts` source runs no discovery, so there is nothing new for the sync to find. `pull-force` does not run this sync at all; it is a destructive recovery command, not a lifecycle path the sync is wired into.

`-force-protocol {ssh,https}` rewrites every repo’s remote to that protocol before pulling, persisting the change (`git remote set-url`, a no-op when the URL already matches) rather than a one-off override — the switch sticks for every command after this one too, the same way a repo’s protocol at clone time sticks for everything downstream of it. Same meaning as `initialise/bootstrap`’s `-force-protocol`, applied to an already-cloned tree instead of at clone time — useful when a repo was adopted rather than cloned by `ComplexGitSync` (`initialise` never touches an adopted root’s remote) and its remote’s protocol does not match what the ambient environment can actually authenticate with. On a failure that looks like an SSH or HTTPS auth problem, the error gains an actionable hint naming `-force-protocol <the other one>` even without the flag.

The rewrite reads the repository’s *actual current* remote URL and only swaps its scheme — `ssh (user@host:path)` to/from `https/http (scheme://host/path)` — keeping the host and path exactly as they already are. It does *not* rebuild the URL from the repository’s stored identity fields (`gitprovider/project_owner_name/...`): those can be missing or stale for a repository loaded from an older `.gts` snapshot (see the “Formal `.gts` snapshot specification” section of the architecture reference), and rebuilding from a wrong or default-GitHub provider would silently aim the pull at the wrong host. Converting the URL that is already configured never has that failure mode. A remote in neither recognised form (a bare local filesystem path, for instance) cannot have its protocol converted; `-force-protocol` then fails with an explicit error naming the repository, rather than guessing.

5.3.9 pull-force

```
$ pixi run cgitssync pull-force [spec.cgs|snapshot.gts] [--force-protocol {ssh,https}]
```

Destructively resynchronises the same tree. Every repository — root, parent, and leaf alike — is forced to the selected remote branch, parent-first, with `git fetch`, `git checkout -B <branch> FETCH_HEAD`, and `git clean -fd`. This command can discard local uncommitted changes and untracked files; it never force-pushes. `-force-protocol` behaves exactly as described under `pull` above.

5.3.10 clone

```
$ pixi run cgitssync clone <spec.cgs> [--target-dir DIR] [--output-path DIR]
```

Clones the full repository tree declared by the `.cgs` file. This CLI command delegates to `ComplexGitSyncClient.clone`. When `-target-dir` is omitted, the project root is derived from the specification. `-output-path` provides the parent directory used to derive the project root.

For each repository, `cgitssync` tries the repo target branch first and falls back to the repo fallback branch when the preferred branch does not exist on the remote. The section “Branches in a `.cgs`” sets out which branch each repository targets and why. Nested `.cgs` files are discovered after parent repositories are cloned, so explicit and automatic nested trees are materialised during the same command.

What the clone leaves behind. A clone still downloads one branch — that is what keeps it quick. What it no longer does is *remember* that restriction: it leaves no single-branch fetch refspec in `.git/config`, so a later `fetch` sees the whole remote rather than the one branch cloned. A workspace cloned before this still carries the narrow refspec; the first `pull` or `push` widens it, with no re-clone needed. `status` deliberately does not: it is a read-only command and never rewrites `.git/config`.

5.3.11 checkout

```
$ pixi run cgitsync checkout <branch> [--gts <snapshot.gts>] [--ref-kind
    branch|tag]
```

Checks out a branch or tag across the whole tree, parent-first. Loads the READY registry from `-gts`, then propagates the ref, creates missing local branches, and runs `git checkout` in every repo. On success the tree remains READY and a new `.gts` snapshot is written.

A branch that exists on the remote is joined, not recreated. When the repository already holds a remote-tracking ref for the branch — because some earlier `pull` downloaded it, or another `checkout/branch` fetched it on demand — `cgitsync` creates the local branch *from that ref*, with tracking set. You land on your colleague’s commits. When it has no such ref cached, it asks the remote directly (one `ls-remote`, no full fetch) before deciding: found, it fetches that one branch and joins it the same way; genuinely not found, it creates a new branch where you currently stand, at no extra cost beyond that one lookup.

This is a change of behaviour, and the reason for it is the failure it removes. `cgitsync` used to create the branch at the current HEAD whenever this workspace had not already cached the remote’s ref — so typing a colleague’s branch name, on a clone that had never fetched it, gave you a branch of that name holding commits that had nothing to do with theirs, and nothing in the output said so. Depending on some earlier command (a `pull`, or a differently-built `cgitsync` during `bootstrap`) to have fetched every branch first was the actual defect: one command’s gap silently became another command’s fork.

`pull` still fetches every branch of each remote before pulling your own, and remains the cheapest way to bring many refs across at once. `checkout` and `branch` no longer depend on it having run first: each fetches the one ref it needs, exactly when it needs it.

5.3.12 close-branch

```
$ pixi run cgitsync close-branch <branch> [--private] [--gts <snapshot.
    gts>]
```

Renames `<branch>` to `closed/<branch>`, tree-wide, leaf-first — locally and on the remote. Requires READY; the tree remains READY on success. Renames only: nothing is deleted, so the commits stay exactly as reachable as before, under a name that says what happened to them. Per repository: the branch is pushed under its closed name first, the old remote name is removed only once the commits are reachable under the new one, and only then is the local branch renamed — the same order `memory reboot`’s own branch archiving already uses. A repository whose remote never had this branch published is renamed locally only, with nothing to remove remotely.

Refuses, before touching any repository, in two cases: closing the project’s own default branch (every repository with no branch of its own falls back to it, so removing it would break that fallback), and closing a branch that any repository in scope is currently checked out on. The second refusal names which repository and stops there — this command closes a branch, it does not move the tree off one; check it out elsewhere first, then close it. A repository with no

local branch named **<branch>** is skipped, not an error: a branch that only ever existed on some repositories (a private/local mount, say) is ordinary, not a mistake.

-private selects the writable configuration repositories instead of the project's own, the same as **branch** (create).

5.3.13 add

```
$ pixi run cgit sync add [PATH ...] [--gts <snapshot.gts>] [--dry-run]
```

With no **PATH** arguments (the default), stages all changes across the tree (**git add -all**), leaf-first. Requires **READY**; tree stays **READY**. With **-dry-run**, **cgit sync** prints the leaf-first execution plan and does not mutate repositories.

With one or more **PATH** arguments, each is resolved (relative to **CGSHOME**, or absolute) to the one repo in the tree that owns it and staged there individually (**git add - <path>**), leaving every other repo untouched. A path outside every repo in the tree raises an error immediately, before anything is staged.

5.3.14 rm

```
$ pixi run cgit sync rm PATH [PATH ...] [--gts <snapshot.gts>] [--dry-run]
```

Removes one or more tracked files, each resolved the same way as **add**'s targeted form: to the one repo in the tree that owns it, where it is deleted from disk and the removal staged (**git rm - <path>**). A plain tracked file only — a path that resolves to a directory, or that does not exist, raises an error rather than partially applying, and a path outside every repo in the tree raises before anything is removed. Requires **READY**; tree stays **READY**. With **-dry-run**, **cgit sync** prints the plan without mutating repositories.

Distinct from and unrelated to **import-submodules**' internal use of **git rm -cached** (index-only, dropping a gitlink while keeping the child's working tree) — **rm** deletes the file from disk.

5.3.15 commit

```
$ pixi run cgit sync commit <message> [--gts <snapshot.gts>] [--no-stage]
  [--dry-run]
```

Commits changes across the tree, leaf-first. Loads the **READY** registry from **-gts**. Repositories with no staged changes after the optional **git add -all** step (suppressed by **-no-stage**) are silently skipped. Requires a **READY** tree; the tree remains **READY** on success. Commit preflight now emits warnings for actionable workspace issues such as dirty worktrees, ahead branches, or stale recorded **commit_sha** values. With **-dry-run**, **cgit sync** prints the planned stage/commit sequence without performing git writes.

The commit message conventionally ends with **CGS#VERSION**.

5.3.16 push

```
$ pixi run cgit sync push [--gts <snapshot.gts>] [--dry-run] [--force-
  protocol {ssh,https}]
```

Pushes all repositories to their configured remotes, leaf-first. Loads the **READY** registry from **-gts**. Each repo is pushed to **entry.remote_name** (defaulting to "origin") on its currently resolved branch. Refreshes the stored hash in **GitTree**. Requires a **READY** tree; the tree remains **READY** on success. Push preflight blocks detached **HEADs**, missing remotes, unresolved merges,

and behind/diverged branches; it warns when the local branch is ahead or when the loaded snapshot records an older `commit_sha`. With `-dry-run`, `cgitsync` prints the push plan without mutating any repository. `-force-protocol` behaves exactly as described under `pull` above.

Folds this project’s own memory first, when one is mounted. Before pushing anything else, `push` folds whatever has accumulated in `.cgitsync` into `.cgitsync/.memory` and sends it — the same work `cgitsync memory push` does by hand, run automatically the moment a command was already about to reach a remote. This happens regardless of `-private`: a memory, once adopted, settles exactly as much as an ordinary push. Nothing happens when the tree declares no memory at all. A memory that is declared but not yet adopted, or one that cannot reach its own remote for any reason, prints a warning and does not block the push — the memory’s own trouble is never a reason to fail somebody else’s publish. `-dry-run` lists `cgitsync memory push` first in its plan under the same condition the real run would perform it.

5.3.17 freeze-release

```
$ pixi run cgitsync freeze-release <name> <message> [--gts <snapshot.gts>]
  [--dry-run]
  [--force-protocol {ssh,https}]
$ pixi run cgitsync freeze-release-force <name> <message> [--gts <
  snapshot.gts>] [--dry-run]
  [--force-protocol {ssh,https}]
```

Minimalist release workflow from a READY tree. It chains public operations in order: `add -> commit -> pull -> push -> freeze`. The force variant uses `pull-force` instead of `pull`. The release name becomes the shared tag and freeze snapshot label; the message is used for the release commit.

The `pull/pull-force` step is skipped (not attempted) when the current branch has no upstream configured yet — e.g. a branch just created and checked out this same session, never pushed before. There is nothing to pull in that case; `push` (already upstream-aware, adding `-u` automatically the same way plain `git push -u` would) publishes the branch on its own. On any branch that already has an upstream, the workflow is unchanged.

`-force-protocol` is forwarded to the `pull/pull-force` and `push` steps above, behaving exactly as described under `pull`; the remote rewrite it makes persists, so the final `freeze` step’s own tag push picks it up too without needing the flag itself.

Unlike a bare `freeze`, this records a `release` row on the ledger entry it writes: the installed package’s own SemVer and build counter, plus `<name>` as the tag actually applied. Name the release `v<semver>` (for example `v3.1.0`) to keep the tag and the recorded SemVer in agreement.

5.3.18 freeze

```
$ pixi run cgitsync freeze <name> [--gts <snapshot.gts>] [--dry-run]
```

Performs a release freeze across all repos, leaf-first: stage/commit, create+push the shared release tag, refresh SHA/state, and write a `.gts` snapshot. Loads the READY registry from `-gts`. Requires READY and leaves the tree READY. Freeze preflight checks remote availability, branch alignment, detached HEADs, unresolved merges, and tag absence. Dirty worktrees and stale recorded `commit_sha` values are reported as warnings because freeze will stage, commit, and snapshot them. Freeze snapshots also include a deterministic `freeze_manifest` section that records freeze invariants (immutable snapshot, validated workspace, synchronized tag, `release-name`, ledger checkpoint) and declares `launch_state` as the canonical restore operation. Their immutable snapshot filename includes the release, for example `gts-000001-release-2026.05.gts`,

while the ledger id remains `gts-000001`. With `-dry-run`, `cgitsync` prints the planned add/-commit/tag/push graph for the loaded workspace without mutating it.

Folds this project’s own memory first, same as push. `freeze` (and, through it, `tag`, which folds the same way on its own combined commit-tag-push) crosses the `.cgitsync` → `.cgitsync/.memory` frontier before doing anything else, under the same conditions `push` does. `freeze-release` folds twice in one run — once via its own `push` step, once via its own `freeze` step — which is harmless: a fold with nothing new pending commits nothing and only re-sends the branch.

5.3.19 import-submodules

```
$ pixi run cgitsync import-submodules REPO_ROOT [--apply] [--recursive]
```

Reads `REPO_ROOT/.gitmodules` and reports (or converts) every declared git submodule to a plain ComplexGitSync nested clone. This is the whole job: turning gitlinks into plain clones on disk. It does not author a `.cgs` — `.gitmodules` never records the root’s own identity, and a checkout worth importing already has a `.cgs` (hand-authored) or can get one from `discover` (§5 “discover”), before or after applying.

Without `-apply` (the default), the command performs a *dry run*: it prints each submodule’s name, path, URL, and branch, without touching the repository.

`-recursive` (same name and meaning as `git submodule update`’s own flag) also converts any submodule that itself has its own checked-out `.gitmodules`, at any depth — root first, not leaf first: converting a submodule stages changes in its own working tree, and converting a deeper level first would make the shallower level’s own preflight check (is the submodule’s working tree clean?) see that staged dirt and reject its own conversion; converting the shallower level first avoids this, since `git rm -cached <path>` only ever touches the parent’s own index, never the submodule’s working tree. A submodule path that was never checked out (`git submodule update -init` not run for it) has no `.git` to read a nested `.gitmodules` from and is invisible either way — the same ceiling `discover` already has for itself. Without `-recursive`, behavior is unchanged: exactly `REPO_ROOT/.gitmodules`, one level.

The report lists every level together. Each entry’s `path` is counted from `REPO_ROOT`, and declared in names the `.gitmodules` file it was read from. A submodule’s own `.gitmodules` path means nothing on its own once more than one repository is involved — in `cawaqsviz`, two of the three submodules are called `docs/...` by their own repository.

With `-apply`:

- Verifies every submodule’s working tree is clean (`git status -porcelain` must be empty apart from that repository’s own `.gitignore`); rejects dirty trees immediately. The `.gitignore` exemption exists because `cgitsync` writes that one file itself in every repository holding a child, so it is routinely dirty in exactly the tree being converted; no conversion can harm it, since `git rm -cached` never touches the child’s working tree.
- Runs `git rm -cached <path>` per submodule, dropping the gitlink from the index while leaving the child’s working tree and `.git` directory intact (no re-clone, no local history lost).
- Removes the converted stanza from `.gitmodules` (deletes the file entirely when all stanzas are converted) and stages the change.
- Appends `<path>` to the parent’s `.gitignore` using the same helper that the `.gitignore` lifecycle sync uses.

After applying, review and commit the staged changes manually:

```
$ git -C REPO_ROOT status          # deleted .gitmodules, modified .
    gitignore
$ git -C REPO_ROOT commit -m "chore: retire git submodules"
```

Note: ComplexGitSync stores no credentials and has no private-repo authentication story. If a submodule’s upstream is private and the ambient environment lacks access, the subsequent `cgitsync initialise` or `pull` will fail at the clone step. The `import-submodules` step itself (which only touches the local index and working tree) will still succeed.

5.3.20 `init-from-submodules`

```
$ pixi run cgitsync init-from-submodules REPO_ROOT [--cgs FILE]
    [--max-depth N] [--dry-run] [--force] [--force-protocol {ssh,https
    }]
```

Adopts a submodule-based checkout in one command: `discover` the repositories under `REPO_ROOT`, write the drafted `.cgs`, `initialise` the tree in place, then convert every submodule with `import-submodules -recursive -apply`. `REPO_ROOT` is a project cloned and `git submodule update -init -recursive`’d by hand; the command performs no clone of the root itself.

The conversion must come last, and this command is the reason why. `initialise` adopts the root in place but deletes and re-clones every *other* repository straight from its remote, and those remotes still declare submodules — so a conversion run first is undone for every repository except the root. Nor can a second pass repair that: `import-submodules -recursive` follows the submodule graph declared by the root’s own `.gitmodules`, which the first pass removed, so no deeper level stays reachable. Running the two commands in the other order by hand produces a half-converted tree that looks finished.

Three conditions are checked before anything is written, so a run that cannot work fails while the checkout is still untouched:

- the discovery resolved at least one repository (skipped when `-cgs` supplies a file, which is then the authority);
- the project name matches the directory name — `CGSHOME` is resolved as `<parent>/<project-name>`, so a mismatch would send `initialise` to a sibling directory that does not exist;
- `REPO_ROOT` has a `.gitmodules`. Without one there is nothing to convert, while the clone step would still delete and re-clone every non-root repository, discarding any uncommitted work in them. `-force` overrides this one.

`-dry-run` prints the discovery and the submodules that would be converted without writing a `.cgs`, cloning, or touching any repository. `-cgs FILE` uses an existing `.cgs` instead of writing one (or writes the draft there, when `FILE` does not exist). `-max-depth N` bounds the discovery step, which is otherwise unbounded — most trees never need it. `-force-protocol` applies to the clone step only; discovery reads each repository’s configured remote as it is.

The command stops at a `READY` tree with the conversion staged but not committed — it touches every repository that held a submodule, and some of those may belong to other people — and prints the `branch/checkout/add/commit` sequence to run next.

5.3.21 `memory`

```
$ pixi run cgitssync memory status [--search-dir DIR]
$ pixi run cgitssync memory list    [--search-dir DIR]
$ pixi run cgitssync memory show <state> [--full] [--search-dir DIR]
$ pixi run cgitssync memory explore [--branch NAME] [--timeline] [--
  search-dir DIR]
```

Looks at what this workspace remembers. A **State** is one recorded snapshot of the tree, named by its own content; the **ledger** holds one entry per operation, saying when that State was seen and by which tool versions.

- **status** — how many States, how long the chain is, when it was last written, which of **verify**'s four answers this memory is in, and the toolchain the records carry. The first entry's versions are shown beside the latest when they differ, so a chain spanning an upgrade says where the upgrade fell.
- **list** — every State, newest recording first, with the commands that recorded it. A State with no timestamp is one no entry records: history from before the ledger, or a file that arrived some other way.
- **show** — one State, every entry naming it, and what each **commit** wrote: the message, the repository, and whether the commit has been published. The argument is the State's content hash or any unambiguous prefix of it; nobody retypes sixty-four characters. **-full** prints whole messages instead of first lines, and the remote and ref each publication went to.
- **explore** — a memory a person can read without a hash. The default view is one row per commit this memory recorded as published, newest push first, on the branch checked out here — what a colleague pulling this branch would see. **-timeline** reads the ledger straight through instead: one row per entry, in the order it happened, so **checkout/merge/push** appear beside **commit** rather than being dropped. **-branch NAME** asks for a different memory branch; only the one actually checked out here can be read this way today, and a mismatch names **memory clone -branch NAME** rather than answering for the wrong branch.

A commit marked **unpushed** exists on this machine and nowhere else. That distinction is the reason the messages are kept at all: while the repository is present **git log** is the better answer, and this record is for when it is not — a deleted branch, an archived project, a fork that went away. The ledger entry that recorded a set of messages carries their fingerprint, so editing one afterwards is a finding **verify** reports.

status, **list**, **show** and **explore** are read-only; nothing they do sends a memory anywhere.

Keeping a memory when the disk does not. A memory can be a repository of its own, mounted at **.cgitsync/.memory** exactly as **.localSpec** is mounted: one repository holding every project's memory, on a branch per project. **.cgitsync** itself stays the workspace's own live state — States, the ledger, logs — so the mount one level inside it is checked out and merged like any other private repository, never blocked by activity that is not its own. Getting there is five commands, run once per project; **tutorials/05_memory.md** walks all five in order.

```
$ pixi run cgitssync repo create github:you/.memory
$ pixi run cgitssync memory mount [--cgs FILE]
$ pixi run cgitssync memory adopt
$ pixi run cgitssync memory push
$ pixi run cgitssync memory branch --project-branch main
```

- **repo create** — creates the repository by running the provider’s own tool (**gh**, **glab**, **tea**), which you have already signed in to. ComplexGitSync stores no credential and sends none. A repository that already exists is an ordinary success; a tool that is missing or signed out prints the command to run and exits 2.
- **memory mount** — adds one entry to a **.cgs** that already exists, leaving the rest of the file, comments included, exactly as it was. Adding it twice changes nothing.
- **memory adopt** — creates **.cgitsync/.memory** fresh and turns it into that repository, on this project branch’s memory branch. It starts empty: the first **memory push** is what folds in whatever **.cgitsync** already held. **-reboot** adopts the same repository identity but skips carrying forward whatever the fallback branch already holds, so the mount’s content starts fresh instead (§5.3.21).
- **memory branch** — creates the memory branch another project branch will need, and pushes it. Needed once, for a memory born on a feature branch: merging **memory-dev** into **main** asks for a branch that has never existed. A merge that finds one missing names this command rather than creating the branch itself, because a branch that is not there is as likely to be a typing mistake as a new branch.

Migrating a memory mounted before memory migrate existed. A memory adopted before this command existed sits directly at **.cgitsync**, sharing it with the workspace’s own live state. **cgitsync memory migrate [-cgs FILE]** moves it onto the layout above — **.git** and every tracked file travel down into **.cgitsync/.memory**, history unchanged; whatever was never tracked stays exactly where it already was, which is where the new layout wants it anyway — and updates the **.cgs** entry that declares the mount to match. A fresh **memory adopt** never needs this: it mounts at the new layout directly.

```
$ pixi run cgitsync memory init      # the .cgs entry to add, and the
    branch
$ pixi run cgitsync memory clone     # bring a memory onto a new machine
$ pixi run cgitsync memory push      # commit what it gained, and push it
```

init proposes and stops: ComplexGitSync speaks Git and nothing else, so it prints the command that creates the repository and waits. Nothing is ever pushed automatically — a machine with no network keeps a complete, verifiable memory and sends it later.

clone refuses to write over a memory that is already here, because a memory nobody has pushed is the only copy of itself.

Starting a memory’s history over. A project’s shape changes — repositories added, removed, restructured — and a memory built for the old shape stops being a clean answer to "what does this project look like." **cgitsync memory reboot** closes the current chapter and opens an empty one, without losing the old one:

```
$ pixi run cgitsync memory reboot
```

In order: whatever **.cgitsync** is holding pending is folded in and pushed under the branch’s current name, exactly as **memory push** already does; the tree’s current shape is exported — from the loaded **.gts**, never a hand-authored file — to a permanent, versioned **.cgitsync/.memory/.cgs/<project>** (N incrementing once per reboot, never overwritten, never reused); the branch is archived, renamed both locally and on origin to **<branch>.archived-<date>**, the old name removed from origin only *after* the new one already holds the same commits, so nothing the memory ever recorded is unreachable for even a moment; and a fresh, empty branch takes the original name, so nothing about how the memory is mounted changes. Nothing is ever force-pushed or deleted. **cgitsync memory adopt -reboot** is the same fresh start for a mount being adopted for the

very first time. Appending — the ordinary `memory adopt`, and the only thing `memory push` ever does — stays the default either way.

What a memory carries off your machine. One path: the tree’s own root, with `$HOME` substituted. Every other path it records — each repository’s, and any path in a command line it logged — is written against the tree as `$CGSTREE/...`, so neither your directory layout nor your user name travels with it.

5.3.22 verify

```
$ pixi run cgitsync verify [--search-dir DIR] [--repair]
```

Says which of five things is true of this workspace’s recorded history. One of them, never a blur of two:

Answer	Exit	Means
verified	0	A chain was read and every link held.
no-history	0	Nothing has been recorded here yet. Not a failure — a new workspace is not a broken one.
legacy	1	History exists, in the single-file <code>.lgr</code> register, which carries no chain. Readable, not verifiable: an edit to it leaves no trace.
corrupt	1	A chain was read and it does not hold.
time-inconsistent	1	The chain held, but its own timestamps move backwards somewhere. Your history is intact; the clock that stamped it was not.

`legacy` exits non-zero on purpose. The question is “is this history intact?”, and “I cannot tell” is not a yes: a build that gates on `verify` must not pass because the evidence is in a format nobody can check. Every workspace created before the hash-chained register is written is in exactly that state.

`time-inconsistent` is deliberately not `corrupt`. Every link checked out — nothing was rewritten — so the two answers send you to look at different things. A clock corrected mid-session, a restored virtual machine, or a machine that disagreed about the hour all read like this, and so does an entry somebody backdated: a date cannot be moved backwards without contradicting the chain around it. It still exits non-zero, because something is wrong even though your history is not.

When there *is* a chain, what is checked: every entry’s `prev` must match the previous entry’s `entry_hash` (`BROKEN_LINK`), every entry’s own `entry_hash` must match its recomputed canonical hash (`BAD_ENTRY_HASH`), `seq` numbers must be gap-free and unique (`SEQ_GAP/SEQ_DUPLICATE`), and the cached `HEAD` must agree with the recomputed head (`HEAD_STALE`).

`CGSHOME` is located the same way as `status`: from `-search-dir`, else `$CGSHOME`, else the current working directory’s ancestors.

With `-repair`, a stale `HEAD` cache is corrected to match the recomputed true head. **Entries themselves are never rewritten or deleted** by `verify`, repair or not — a broken chain is reported, not silently healed, since a register that can be edited back to looking clean provides no evidence of anything.

The States on disk are checked too. Beyond the chain itself, `verify` cross-references what the entries say against what is there: an entry naming a State that is missing (`MISSING_STATE`), a stored snapshot whose content no longer hashes to the name it is filed under (`STATE_DIGEST_MISMATCH`),

and a State on disk that no entry records (`ORPHAN_STATE`). The second is what naming a State by its content bought: a stored snapshot can no longer be edited quietly.

Where the history lives. `.cgitsync/lgr/`, one file per entry, appended to by every command that writes a State. Each entry records the versions that produced it — `cgitsync`, `git`, `pixi`, and the data backends `dvc` and `git-lfs` — inside the entry hash, so an edited version string is as detectable as an edited command. The older single-file `.lgr` register is still read, and is no longer written.

5.3.23 `env` and `env check`

```
$ pixi run cgitsync env [--search-dir DIR]
$ pixi run cgitsync env check [--search-dir DIR] [--cgs FILE]
```

`env` observes the environment that can make the current tree usable. It reports the machine architecture, Pixi platform, operating system, kernel and libc; raw and parsed versions for Python and the tools `ComplexGitSync` drives; provider authentication as a fact; and the paths and digests of recognised manifests in the tree. A missing tool is printed as `none`, never as an empty value. Tokens, keys, user names, key paths, and paths outside `$CGSTREE` are never part of an Environment record.

`env check` compares that observation with requirements from the loaded tree's source `.cgs`. Pass `-cgs FILE` to check an explicit authoring document instead. Missing tools and versions older than the declared minimum are drift: the command names each mismatch and exits non-zero, which makes it suitable for continuous integration. Extra observed tools are reported as information and do not make the check fail.

The check is separate from tree operations on purpose. `initialise` and `pull` may warn about drift, but a mismatch never blocks them. Both environment commands are read-only and install nothing.

The optional authoring declaration names the repository where the environment is installed and requirements that cannot be inferred from a lock file:

```
environment_root = "my-project"

[environment]
tools = { git = "2.43", pixi = "0.66" }
compilers = ["cc"]
system_libraries = ["libssl"]
services = ["postgresql"]
manifests = ["Cargo.lock"]
```

The root must name a configured repository (or `root`). Manifest contents remain in Git; the Environment record stores only tree-relative paths and digests.

5.3.24 `launch-release`

```
$ pixi run cgitsync launch-release <name> [--gts <snapshot.gts>]
```

Checks out the frozen release tag `<name>` across the loaded `READY` tree, parent-first, and writes a fresh `.gts` snapshot. This is the release oriented CLI path for returning a workspace to a frozen version; direct tag creation remains available from the Python client API as `tag()`.

The integration suite validates local no-network lifecycle coverage through the CLI path, including initialisation and the `READY` git command cycle `add -> commit -> push -> freeze -> launch-release`.

5.3.25 view-tree

```
$ pixi run cgitSync view-tree [spec.cgs|snapshot.gts] [--depth N] [--collapse REPO]
```

Renders a deterministic terminal tree that focuses on repository topology and operational state. Each line reads:

```
name (node_type) [sync_state] @sha br=<branch> [fb=<fallback>]
```

br= is printed for *every* repository, including the common case where the branch is simply the tree's default, so that reading **view-tree** tells you the branch each repository targets without opening the **.cgs**. **fb=** appears only when the declared fallback branch differs from that target, since a fallback equal to the target says nothing. Use **-depth** to limit visible levels and repeat **-collapse** to hide selected subtrees.

On a tree with private mounts, **br=** is where you see pinning at work: the unprivate repositories carry the branch you moved the tree to, and each private mount still shows its own.

Branch Topology Propagation Rules (T35):

1. **Reference branch:** The root repository's current branch is the canonical reference for all repos.
2. **Leaf-to-root inheritance:** Branch targeting flows root-first via `propagate_global_branch`, verified on-disk by `ComplexGitSyncClient.validate_topology()` (Python API only) without issuing any git writes.
3. **Allowed divergence:** Repositories in a frozen (**TAG**) state produce a `tag_divergence` diagnostic that does *not* make the topology incoherent.
4. **Incoherent states** (blocking): `misaligned_branch` (different branch from root) and `detached_head` (detached HEAD without a tag reference).

5.3.26 What cgitSync promises a script

Exit codes. Three, meaning the same thing in every command:

Code	Means
0	The command did what was asked.
1	It ran, and the answer is no — a merge conflict, a tree that is not READY , a verification that found something.
2	It could not run — bad arguments, no workspace, a missing or unreadable file.

The distinction that matters to a script is between “I asked and the answer is no” and “I could not ask”. **validate** is the one command that judges a document rather than acting on one, so an invalid document is its answer (1); every other command exits 2 on the same document because it could not run at all.

An expected failure prints one line on `stderr` and no traceback. A traceback is therefore a defect report, not a usage error — which is the point of mapping the expected ones.

-json. **status** and **verify** accept it. Each prints one JSON object on `stdout` and nothing else; every human-facing line goes to `stderr` instead, and the exit code is unchanged, so a caller may use either signal. A failure prints one object too, carrying `"status": "error"` and the exit code, rather than leaving a pipe with nothing to parse. **-help**, **-version** and a command line that fails to parse are deliberately not JSON: the flag itself may be what failed.

The object carries `schema_version`. The promise attached to it is **additive only**: new fields may appear, existing fields do not change meaning and do not vanish.

5.3.27 Where a command works: the default workspace

Every command that is not given an explicit path finds its workspace from three inputs, in this order: `-search-dir`, then `$CGSHOME`, then the ancestors of the current directory. When none of them lands on a `.cgitsync` directory there is a fourth answer: an empty workspace of ComplexGitSync’s own, under `$HOME/.cgs` (or `$CGSPATH` when that is set).

That workspace is a real directory holding a valid `.gts` with no repositories in it. An empty tree is not an error — it is a project that has not started — so `status` answers `no living project yet`, names the three commands that start a project, and exits 0.

- **Created once, then reused.** Its path is recorded in `$HOME/.cgs/default`; later runs read that pointer instead of minting a second empty workspace.
- **It never chooses for you.** When other workspaces exist under the root they are listed, each with the `export CGSHOME=...` line that would select it, and none is selected automatically.
- **It never overrides `-search-dir`.** A directory you named that holds no workspace is an error, not an invitation to work somewhere else.

The `use_case` field. Every discovery line also says which of the two ways of running is in force: `nested` when the ComplexGitSync being executed lives inside the workspace it manages, `standalone` otherwise. It is observed and printed, never obeyed — nothing behaves differently because of it. It exists so that a user who believes they are in one case while standing in the other is told.

5.3.28 `status`

```
$ pixi run cgitsync status [--gts FILE]
```

Summarises tree readiness and per-repository sync state. The table reports the scope, local branch, upstream branch, local worktree state, ahead/behind tracking counts, current HEAD, and the commit recorded in the loaded `.gts`.

The `cgitsync_branch` field. The `summary` line carries `cgitsync_branch=<branch>`: the branch the project is on, which is the branch its root repository is on. It is what `cgitsync branch` and `cgitsync checkout` set, and what every other repository follows.

The table shows a branch per repository and they are not all the same on purpose — a `private/local` repository keeps this project’s settings on a branch named after the project, so a project on `apoub` shows `ComplexGitSync_apoub` there. The field is the one place that says which of those branches is the project’s own.

Shown	Means
<code><branch></code>	The branch the root repository is on.
<code>detached</code>	The root is parked on a commit rather than a branch. <code>cgitsync checkout <branch></code> puts the tree back.
<code>unknown</code>	There is no branch to report: no project has been loaded, or Git could not be asked.

The field is printed in every state. A field that disappears when the answer is awkward is harder to read, by eye or by script, than one that says it does not know.

The SCOPE column. It says, in three words, what kind of repository each row is and whether you may write to it:

Shown	Means	In the .cgs
project	this project's own; write freely	no private
private/local	shared, and yours to write	private, writable
private/distant	shared, read-only	private alone

private means the repository is shared with other projects — a configuration repository. **local** then means this project may write to it, normally because it sits on a branch named after your project; **distant** means you only read it. A repository nested inside a private one is shown the same way, because it is just as shared.

commit and **push** reach the **project** rows; **-private** reaches the **private/local** rows; nothing writes to a **private/distant** row. When any private row is present, the table prints a legend saying so.

The SYNC column. It says how the branch a repository is on stands against the branch it tracks — the one named in `UPSTREAM_BRANCH`, which reads - when there is none. There are six values:

Shown	Means
synced	Level with the upstream.
ahead(+N)	N commits here that the remote does not have. push sends them.
behind(-N)	N commits on the remote that are not here. pull fetches them.
diverged(+N/-M)	Both, from a common ancestor. merge or pull-force resolves it.
no-upstream	This branch was never pushed, so there is nothing to compare it to. Normal for a branch you just made, and for a private/local repository that only push -private ever sends.
unknown	The branch names an upstream that does not resolve. pull or push repairs it; if it persists, the remote is unreachable or the ref was deleted.

The last two are worth telling apart. **no-upstream** is a normal state of a branch nobody has pushed yet; **unknown** is a fault. They used to share the one word **unknown**, which made the column not worth reading — an ordinary new branch looked exactly like a broken one.

unmeasured in the summary line. The summary counts the **no-upstream** and **unknown** rows as **unmeasured**, separately from **ahead** and **behind**. Those rows were previously folded into **ahead=0 behind=0**, which reported a measurement nobody had taken. A repository nobody could measure is not a repository that is level, and the summary no longer says the second when it means the first.

The memory mount reading dirty mid-push, or HEAD ending in * right after, is normal. If the tree mounts its own memory at `.cgitsync/.memory` (§5.3.21), that row is an ordinary **private/local** repository the rest of the time — clean before and after every command except **memory push** itself, which folds `.cgitsync`'s own pending content into it before committing. One thing about it is still worth knowing:

- **HEAD ends in * right after memory push.** The * means the loaded `.gts` recorded a different commit than the one actually there now. `memory push` moves the memory’s real commit directly and does not write a new `.gts` itself, so the State most recently written still names the commit from before the push. The very next command that writes a State reads the memory’s true HEAD and the two agree again — it never needs a second `memory push` to catch up, only an ordinary next command.

Not worth chasing to a clean reading on purpose: a memory that never showed this for a moment would not be recording what `memory push` just did.

5.3.29 merge

```
$ pixi run cgitssync merge <branch> [--into TARGET] [--private] [--ff-only|--no-ff] [--dry-run] [--resolve]
```

Merges a branch across the tree, leaf-first. The argument is always the **project’s** branch; each repository works out what that means for itself. A project-owned repository merges that branch. A private/local repository merges the branch derived from it (`<base>_<branch>`), because that is where its settings for that project branch live — it has no branch of the project’s own name at all.

-into: merging into a branch you are not on. Without it, `merge` merges into whatever is checked out, so reaching another branch means running `checkout` first. `-into TARGET` names the target and does both halves in one command.

That is not a convenience when the tree being merged **contains the ComplexGitSync you are running** — the developer checkout, which installs itself editable. A separate `checkout` replaces that build, and the merge typed after it runs under the older one, against a workspace the newer one wrote. A single command finishes under the build it started with, because Python has already imported its modules.

Every repository is checked before any is touched, so a conflict or a missing target leaves the whole tree where it was: still on the source branch, nothing checked out and nothing merged. A fast-forward is reported as one, which is usually why a repository looks untouched afterwards. `-into` never creates its target: a branch that is not there is as likely to be a typing mistake as a new branch, so it is named and the command stops. It cannot be combined with `-resolve`, which promises the opposite about a conflict.

`checkout` warns when it is about to install a different build of the tool, and still performs the checkout — looking at an older branch is legitimate, being surprised by it is not.

A `-private` or `-all -into` call only ever touches part of the tree, and the rest is meant to still be somewhere else until it too is asked to move. So a second scoped call — `-private` after the plain form, or `-all` after either — finishes what the first one left behind, rather than refusing those repositories for not having moved yet.

`-private` merges the writable configuration repositories instead of the project’s own. Read-only ones are never merged.

A conflict in any repository leaves no repository merged.

Every repository in scope is checked before any is merged, using a read-only test that touches neither worktree nor index. A tree-wide merge that stopped halfway would leave a workspace no `.gts` describes and no command undoes, so it is prevented rather than reported. `-dry-run` prints the plan — and, importantly, the branch each repository would actually merge, and whether it would merge at all.

The refusal names the files. When `cgitsync` refuses, it names every blocked repository and, under each, every file that conflicts:

```
merge refused; no repository was merged: ComplexGitSync: tests/unit/
  test_documents.py
```

A file that both branches changed is only listed when it genuinely conflicts. Binary files count: a tracked PDF changed on both sides conflicts just as a source file does, and is reported the same way. `-dry-run` shows the same list without merging.

-resolve: fix the conflict instead of reading about it. `-resolve` merges one repository at a time and stops at the first that conflicts, then opens that repository in your merge tool.

-resolve gives up the guarantee above.

Repositories merged before the conflict stay merged, so the tree can be left partly merged. That is the price of reaching a conflicted worktree at all, and it is why this is opt-in: the command prints the warning before it writes anything, then names what it merged, where it stopped, and which repositories it never reached.

Your own `merge.tool` is used whenever you have configured one. Otherwise VS Code is suggested when it is available, passed for that one call — your Git configuration is never written. With no merge tool available at all, the command prints the command to run by hand and stops, rather than failing.

Check out the target first. `merge` brings a branch *into* the one the tree is on, exactly as `git merge` does. Running `cgitsync merge multi-branch` while the tree is still on `multi-branch` would merge it into itself — which Git reports as success while doing nothing — so `cgitsync` refuses and says to check out the target branch first.

Landing a feature branch is therefore two commands:

```
$ pixi run cgitsync merge multi-branch
$ pixi run cgitsync merge --private multi-branch
```

5.3.30 pull -private

```
$ pixi run cgitsync pull --private
```

The other direction. While a project feature branch is open, work lands on a private/local repository's base branch and its derived branch does not see it. This fetches and merges the base branch into each one, so a settings branch does not drift behind the project's. A repository already on its base branch has nothing to take.

5.4 Document Formats

`.cgs` means **ComplexGitSync** specification, `.gts` means **GitTreeState** snapshot, and `.lgr` means **Local Git Register**.

5.4.1 Local Authoring Spec (.cgs)

A TOML file that describes the repository tree to manage. It is the only hand-authored input; it is never modified at runtime. The normal form has two top-level keys:

```
project = "CGSi11"
repos = [
    "gitlab:CGS_test/CGSi11",
    "gitlab:CGS_test/CGSi12",
    "github:flipoyo/CGSi11",
]
```

Repository identifiers have the form `provider:owner/repository`. Nested GitLab namespaces are supported; the final path segment is the repository name. The document pipeline is:

PARSE -> NORMALIZE -> VALIDATE -> canonical CgsDocument

The canonical providers and hosts are:

Provider	Host	Remote generation
github	github.com	SSH and HTTPS
gitlab	gitlab.com	SSH and HTTPS
codeberg	codeberg.org	SSH and HTTPS
custom	explicit URL required	SSH and HTTPS

Repository-ID parsing is provider-independent and deterministic. For example:

```
codeberg:GX4G/GX4G
```

normalizes to `gitprovider = codeberg`, `project_owner_name = GX4G`, and `repo_name = GX4G`. A custom identifier must use an inline table so the host is explicit:

```
{ repository = "custom:team/tool", gitprovider_url = "https://git.
  example.com" }
```

No host is inferred for `custom`. The configured access protocol then selects `git@host:owner/repository.git` or `https://host/owner/repository.git`.

The textual grammar and its only parser, `parse_repo_id()`, live in `cgs_format.py`. The provider model in `git_repo.py` receives the already-separated provider, owner, and repository fields and composes remotes; it does not parse authoring shorthand.

The `.cgs` format pipeline is deterministic and offline. A syntactically valid repository may therefore reference a host, owner, branch, or tag that is not currently reachable. Remote existence and reference availability are checked only by explicit Git operations during runtime resolution and execution.

Normalization supplies `default_branch = main` (see “Branches in a `.cgs`” below for the full chain), makes `fallback_branch` equal to that default, selects `access_protocol = ssh`, enables `nested_config = auto`, and uses the repository name as `relative_path`. If exactly one repository name matches `project`, its path is inferred as `..`.

Advanced inline tables are used only where a value differs from those defaults:

```
project = "demo"
repos = [
    "github:owner/demo",
    { repository = "github:owner/docs", relative_path = "documentation",
      access_protocol = "https" },
]
```

The legacy advanced `[project]` and `[[repos]]` tables remain accepted for project-wide branch settings, custom provider URLs, tags, and other explicit overrides. Unknown providers, malformed identifiers, duplicates, and missing `project` or `repos` are rejected before a `GitTree` is built.

Two checked-in files document this boundary. Copy `examples/template.cgs` for normal authoring. Its equivalent `examples/normalized_template.cgs` expands every inferred field and is provided as a developer reference for the canonical `CgsDocument` shape. The two files are regression-tested for semantic equivalence.

The repository also keeps three checked-in, didactically ordered tutorials under `tutorials/` — easiest first, see `tutorials/README.md` for the guided order — exercising this authoring surface end to end: `tutorials/01_first_multi_repo_workspace.md` walks a fully exercised local sandbox topology (CGS11) through the complete CLI lifecycle; `tutorials/02_onboarding_a_real_build_tree.md` hand-authors a `.cgs` for the larger, real-world `cawaqs` tree and the point where `cgitsync` hands off to the project’s existing make-based build workflow; and `tutorials/03_adopting_a_real_project.md` covers the real-world `cawaqsviz` project (not synthetic), which has no `.cgs` of its own and still tracks three repositories on two levels as git submodules, as one verified procedure: clone, check out the submodules, adopt the whole tree with `init-from-submodules` (`discover + initialise + import-submodules`, in that order), then `branch/checkout/add/commit/push`.

`examples/cawaqsviz.cgs`, the hand-authored alternative mentioned in `tutorials/03_adopting_a_real_project.md` above, describes <https://gitlab.com/cawaqs/gviz/cawaqsviz> and exercises several patterns above at once: a nested GitLab subgroup path (`cawaqs/gviz/cawaqsviz`, three segments rather than a plain `owner/repository` pair), an explicit `relative_path = "."` on the root entry rather than relying on the project-name auto-mount rule, two GitHub children mounted at non-default relative paths (`external/HydrologicalTwinAlphaSeries` and `docs/CWV_user_guide`). Neither child has a `.cgs` of its own, but no override is needed for that: the default `auto` nested discovery resolves a repository with zero `*.cgs` matches as a normal leaf. This file previously shipped with a nonexistent repository identifier and an invalid `default_branch`; it was corrected and given real end-to-end test coverage (`tests/integration/test_cgsi_topology.py`, and one verified run against the live repositories). See `tutorials/03_adopting_a_real_project.md` for the full worked example, and `bootstrap` above for running `ComplexGitSync` against it without cloning `ComplexGitSync` inside the tree it manages.

Generated specifications use the same concise representation. The `GitTree.to_cgs()` convenience method delegates model projection to `cgs_format.py`; neither `GitTree` nor `GitRepo` formats TOML or implements the authoring grammar. Serialization omits every default that normalization can deterministically reconstruct and retains inline tables only for exceptional settings such as a non-default branch, path, protocol, tag, custom host, or discovery rule.

The supported round trip is semantic:

```
.cgs -> CgsDocument -> GitTree -> CgsDocument -> .cgs
```

Parsing and normalizing the generated file must reproduce the initial canonical document. Whitespace, comments, and TOML table layout need not be byte-for-byte identical.

5.4.2 Branches in a `.cgs`

Four fields decide which branch a repository lands on. Three of them fall back to each other, so a `.cgs` that names no branch at all still sends every repository somewhere.

Field	Where it is written	Default
<code>project.default_branch</code>	the project table	<code>main</code>
<code>repos[].default_branch</code>	one repository entry	the project’s
<code>repos[].fallback_branch</code>	one repository entry	that repository’s own
<code>repos[].private</code>	one repository entry	<code>false</code>
<code>repos[].writable</code>	one repository entry	<code>false</code>

Read the first three as one chain, longest reach first:

```
fallback_branch -> default_branch -> project.default_branch -> main
```

`project.default_branch` is the tree's branch: every repository inherits it unless the repository says otherwise. `repos[].default_branch` is the branch that one repository targets. `repos[].fallback_branch` is used when the target branch does not exist on the remote — `cgitsync` clones the fallback instead and records that it did so, rather than failing.

Each link is silent. Nothing warns you that a decision was made, so **state `project.default_branch` in every `.cgs` you write**, even when the answer is `main`. A reader of the file can then see the branch the tree lands on without knowing what `cgitsync`'s built-in default happens to be. Every `.cgs` shipped with `ComplexGitSync` does this, and a test enforces it.

What private means

A repository is **private** when it is **shared with other projects** and must stay on its own branch.

Branch moves stop at a private repository. Tags do not.

`cgitsync branch <name>` and `cgitsync checkout <name>` skip a private repository and leave it on its declared `default_branch`, because moving it would drag every other project that mounts it onto your branch.

Read-only and writable configuration repositories

A private repository is a **configuration repository**: shared with other projects rather than owned by this one. `private = true` alone means **read-only** — `cgitsync` will not write to it. Add `writable = true` for the case where the repository lives outside your project but the part you use is yours, normally because it sits on a branch named after your project that nobody else reads:

```
{ repository = "github:you/.sharedSpec", default_branch = "main",
  private = true },
{ repository = "github:you/.myNotes", default_branch = "MyProject",
  private = true, writable = true },
```

Read-only is the default because the expensive mistake is writing to something shared by accident, never the reverse. `writable = true` on an unprivate repository is rejected at validation: an unprivate repository is this project's own and is always writable.

This is a **safety rail, not a permission system**. It stops a tree-wide sweep from writing where you did not mean; it cannot stop anyone running `git` in that directory by hand, and it is no substitute for branch protection on the remote.

A private/local repository gets a branch per project branch. A private/distant repository is private to its owner: you read it, only that owner writes to it, and nothing you do moves it — so a branch move leaves it exactly where it is. A private/local repository is the opposite case: it holds settings that configure *your* project and you commit to it, so it needs somewhere to record them per project branch.

`<project name>` on `main`, `<project name>_<branch>` on any other branch

The base is the ****project's**** name. `main` takes no suffix, because the project's main line's settings branch is simply the project's name — which is what every existing tree already has, so nothing has to move. The separator is an underscore because hyphens already occur inside branch names.

A user never types the derived name. `cgitsync checkout <branch>` puts the project’s own repositories on that branch and each private/local one on the branch this rule names, creating it when it is missing.

Without this, one settings branch serves every branch of the project, and documentation written for an unfinished feature is live on the default branch describing code that is not there.

-private everywhere. Every command that touches Git takes it: `clone` aside, that is `pull`, `pull-force`, `checkout`, `branch`, `close-branch`, `add`, `rm`, `commit`, `push`, `merge`, `tag` and `freeze`. It narrows the command to the writable configuration repositories alone, so they can be worked on without reaching for plain `git`. Read-only configuration repositories are never written to, with or without it. Four of those commands also take `-all`, described next.

-all: both halves, one command. Running every write twice — once plainly, once with `-private` — is the right default when the two changes are different changes. When they are one change, `-all` does both halves in a single pass, under a single commit message. Four commands take it: `add`, `commit`, `push` and `merge`.

Form	Reaches
<code>cgitsync <cmd></code>	this project’s own repositories
<code>cgitsync <cmd> -private</code>	the writable configuration repositories
<code>cgitsync <cmd> -all</code>	both of the above, in one pass

The plain form has not moved, so nothing an existing script runs changes meaning. `-all` and `-private` cannot be combined: `-all` already includes everything `-private` selects. No form of either writes to a read-only configuration repository — “all” means every repository `cgitsync` may write to, never every repository in the tree.

The output still names the two halves apart, so giving up the typing does not mean giving up seeing which repository was written as what. On a tree with no writable configuration repository, `-all` does the project half and reports the other as empty; that is the common case, not a fault. An explicit `-private` on the same tree still stops, because there the user asked for something that is not there.

`merge -all` checks every repository before merging any.

That is why `-all` is one pass and never two. Two sequential merges mean two preflights: the project half would merge, the configuration half would then refuse, and the tree would be left half-merged — exactly the state `merge`’s whole-scope check exists to prevent. `-dry-run` previews the same combined scope, and `-resolve` keeps its own warning and its deliberate partial progress.

A pin covers everything inside it. A configuration repository often holds repositories of its own, reached through its nested `.cgs`. Those are just as shared as the repository holding them, so they inherit its state: **a repository inside a read-only configuration repository is read-only, even when its own entry says nothing about it.** One inside a writable configuration repository is writable, and `-private` reaches it.

A nested entry may restrict itself further — writing `private = true` on its own line, with no `writable`, makes it read-only inside a writable parent. It cannot go the other way: `writable = true` inside a read-only configuration repository is ignored, because a repository can never be more open than the one that contains it.

Which repositories each command touches

Command	Reaches	Why
add, commit, push	this project's own	they write this project's history
the same, -private	writable config repos	shared, but on your own branch
branch, checkout	this project's own	a branch must not spread
tag, freeze-release	everything writable	they create <i>and push</i> a tag
clone, pull, status	everything	reading a shared repo is the point

Preflight follows the same scope. Each command checks only the repositories it is about to touch. A configuration repository that is behind its upstream, or sitting on its own branch, never blocks a `commit` of this project's own work — that repository is not part of the commit. Under `-private` it is in scope, and its branch is then checked against the branch it is *private* to, not against the one the project is on.

`-private` is exclusive, not additive: it selects *only* the writable configuration repositories, so a shared repository gets its own command and its own commit message rather than being swept into this project's. Asking for it in a tree with no writable configuration repository is an error naming the read-only ones, not a silent no-op. Every scoped command reports what it left alone:

```
scope=project skipped=5 configuration repo(s) (.claude, .localSpec with
--private)
```

`tag` and `freeze-release` stop at read-only repositories because they push. Nothing is lost: a `.gts` snapshot records every repository's exact `commit_sha`, read-only ones included, so an exact tree is rebuilt from the snapshot rather than from tags.

In practice shared config mounts are usually dot-named (`.agentSpec`, `.localSpec`), and `cgitsync discover` drafts a dot-named repository with `private = true` for that reason. The rule is *shared*, not *hidden*, though: a plainly named repository that two projects mount should be private too.

The branch a repository is private to decides who sees your commits. A mount private to a branch named after your own project is yours to push freely. A mount private to `main` is read by every project that mounts it, so a commit there is published the moment it is pushed — no branch, no pull request, no review. Check which of the two you are looking at before committing to a private mount.

The two meanings of “force”

`pull-force` is destructive: it discards local changes. `-force-protocol` is not: it only switches a remote between `ssh` and `https`. The two are unrelated despite the shared word.

5.4.3 Repository Placement and Nested Discovery

For every non-root entry, `relative_path` is relative to the parent repository represented by the current `.cgs`; it is not relative to the process working directory. At the top level that parent is the project root. While expanding a nested config, it is the repository containing that config. The default is the repository name. A unique repository matching the project name uses `.` because it represents the current root.

The `nested_config` option controls whether traversal continues after the repository is available:

- `auto` (default) loads the sole root-level `*.cgs` file, if one exists; finding none resolves the repository as a normal leaf; multiple matches are rejected as ambiguous.

A consequence worth knowing before you publish a repository: **a repository that other people may mount inside their tree must keep exactly one .cgs at its root**, because `auto` has no way to choose between two. The tree's own root is exempt — it is never scanned for its own `.cgs`, since that would recurse — so a root holding two `.cgs` files works fine until the day someone mounts it. The error names both files and how to resolve it.

- `disabled` does not inspect the repository for nested config.
- A relative path ending in `.cgs` loads that exact file inside the repository, failing if it does not exist. It may not escape the repository root.

For the integration topology, `CGSi12.cgs` contains:

```
{ repository = "github:flipoyo/CGSi12", relative_path = "../CGSi12" }
```

Because the config describes the tree below `CGSi12`, the path resolves from that directory:

```
<CGSHOME>/CGSi12/../../CGSi12 -> <CGSHOME>/CGSi12
```

It therefore references the existing sibling declared by `CGSi11.cgs`, instead of cloning another `CGSi11` under `CGSi12`. Discovery recognizes the already-registered absolute path and keeps the canonical entry — this guard runs before `auto` would ever get a chance to reopen `CGSi11.cgs` through the cross-reference, so no `nested_config` override is needed here.

5.4.4 Git Tree State Snapshot (.gts)

A TOML file generated automatically by bootstrapping and release operations. It must never be hand-edited. It captures the exact commit SHAs and lifecycle state of the tree at a point in time.

Top-level sections:

- `[document]` — `format_version`, `schema_version`, `generated_at`, `command_origin`, `hash_algorithm`, `snapshot_hash`.
- `[project]` — project name and `root_absolute_path`.
- `[tree_state]` — `lifecycle_state`, `is_ready`, `registry_complete`.
- `[[repo_state]]` — one entry per repo; includes `commit_sha`, `sync_state`, and ref information.

Why both version fields and hash metadata exist:

- `format_version` identifies the broad document family and keeps long-range compatibility intent explicit.
- `schema_version` identifies the exact field-level contract used by validation and canonical snapshot serialization.
- `snapshot_hash` (with `hash_algorithm = "sha256"`) provides deterministic workspace identity: identical tree state produces identical digest even if volatile metadata (`generated_at`, `command_origin`) differs.

How this is used at runtime:

- During snapshot generation, `ComplexGitSync` writes `schema_version`, `hash_algorithm`, and canonical `snapshot_hash`.

- During load/validation, `snapshot_hash` is recomputed from the canonical payload and must match.
- The project-local `.lgr` register uses this canonical hash to deduplicate equivalent snapshots.

5.4.5 Local Git Register and Sync Ledger (`.lgr`)

A TOML file maintained per project at `<project-root>/<Project_name>.lgr`. It is updated automatically whenever a `.gts` snapshot is written.

Sections:

- `[register]` — current snapshot pointer (`current_snapshot_id`, `current_snapshot_hash`, `current_snapshot_path`).
- `[[snapshots]]` — list of stable `gts-XXXXXX` identifiers, one per unique workspace state, deduplicated by SHA-256 hash.
- `[[ledger]]` — append-only list of synchronisation events forming a DAG via `parent_sync_ids`.

Ledger event schema:

```
[[ledger]]
sync_id      = "lgr-000001"
parent_sync_ids = []
operation    = "clone"
timestamp    = "2026-05-20T19:48:50.159Z"
actor        = "username"
workspace_hash = "5f7c8a2d..." % 64-character SHA-256 hex digest
gts_snapshot_id = "gts-000001"
affected_repos = ["demo", "dep-a"]
```

The `workspace_hash` field equals the canonical SHA-256 digest (`document.snapshot_hash`) of the associated `.gts` file, creating a direct link between each ledger event and the immutable workspace state it records. Events are parents-before-children in DAG order.

5.5 Worked Examples: From `CGSil1` to `cawaqsviz`

`ComplexGitSync` ships a same-numbered tutorial under `tutorials/` for each level below (`tutorials/README.md` lists them in order). Run through them in order: each one adds exactly one new idea on top of the last, so together they form a guided path from the simplest possible `.cgs` to a real project with no `.cgs` of its own at all.

Levels 2 and 3 have their `.cgs` checked in under `examples/`. Level 1 does not: `CGSil1.cgs` lives in the `CGSil1` repository itself, which is what the transcript below clones.

Every command below is a Pixi task. `cgitsync` is not a globally installed executable — always run `pixi run cgitsync ...`, never a bare `cgitsync ...` (see the §5 “CLI Overview” note above).

Level	Topology	Entry point	New idea introduced
1	CGSil1	hand-authored .cgs	minimal spec, name-matching auto-mount
2	cawaqs	hand-authored .cgs	the same minimal-field style as Level 1, at 19-repo scale
3a	cawaqsviz	hand-authored .cgs	explicit overrides on a real, irregular tree
3b	cawaqsviz	discover	drafting a .cgs from a checkout, no authoring at all
3c	cawaqsviz	import-submodules	migrating an existing git-submodules project

Levels 3a–3c are not three different projects: they are three different *starting points* for the same real project, `cawaqsviz`, chosen so this section can demonstrate every route `ComplexGitSync` offers for arriving at a working `.cgs` — write it by hand, derive it from a checkout, or migrate it from an existing submodules setup — without inventing a fourth toy topology to do it. It is the most advanced level not because `cawaqsviz` is a larger tree than `cawaqs` (it is much smaller, three repos against nineteen) but because, unlike either earlier level, it has no `.cgs` anywhere upstream to copy from.

5.5.1 Level 1 — CGSil1: the minimal synthetic topology

`CGSil1` is the easiest way into `ComplexGitSync`: a 4-repository, mixed-provider tree with every field left at its default. Its full `.cgs` and the rationale for each line are in §5 “Local Authoring Spec” above; `tutorials/01_first_multi_repo_workspace.md` carries the full transcript with expected output for every step. What follows is the command sequence itself.

```
# 1. Fetch the root repo and cgitSync itself (nested mode: cgitSync is
#    cloned inside the tree it will manage).
$ git clone https://gitlab.com/CGS_test/CGSil1
$ cd CGSil1 && git clone https://github.com/flipoyo/ComplexGitSync
$ cd ComplexGitSync

# 2. Check the spec offline, no network beyond parsing.
$ pixi run cgitSync validate ../CGSil1.cgs

# 3. Preview the topology before touching disk.
$ pixi run cgitSync view-tree ../CGSil1.cgs

# 4. Clone every declared repo and reach READY.
$ pixi run cgitSync initialise ../CGSil1.cgs

# 5. The ordinary git cycle, run tree-wide from here on --- no path
#    argument needed, the .gts snapshot is discovered automatically.
$ pixi run cgitSync pull
$ pixi run cgitSync add
$ pixi run cgitSync commit "my commit message"
$ pixi run cgitSync push

# 6. Minimalist release: stage, commit, pull, push, freeze in one call.
$ pixi run cgitSync freeze-release v1.1.0 "release v1.1.0"

# 7. Return to a frozen release at any later point.
$ pixi run cgitSync launch-release v1.1.0
```

The only `.cgs` idea this level relies on is the **name-matching auto-mount**: because `CGSil1`’s repository identifier’s last segment equals `project = "CGSil1"`, the root is inferred

at `relative_path = "."` with no override written anywhere. That convenience is exactly what stops being safe once a project’s identifier and display name diverge — which is where Level 3a picks up.

5.5.2 Level 2 — cawaqs: the same minimal style, at scale

`cawaqs` scales Level 1’s minimal, no-override authoring style up to a real 19-repository build tree (root + 17 physics libraries across five GitLab groups + one required build-tooling repo), instead of a synthetic 4-repo one. Every entry’s on-disk layout already matches the default `relative_path = <repo_name>`, so not a single `relative_path` override is written anywhere in `examples/cawaqs.cgs` — only the shared branch-fallback convention. Full narrative in `tutorials/02_onboarding_a_real_build_tree.md`.

```
$ pixi run cgitssync validate examples/cawaqs.cgs
$ pixi run cgitssync bootstrap examples/cawaqs.cgs cawaqs-smoke-test
$ pixi run cgitssync view-tree --search-dir <path bootstrap printed>

# Move the whole 19-repo tree to a shared branch in one call, falling
# back to main per-repo wherever that branch does not exist.
$ pixi run cgitssync checkout my-feature-branch
```

Reaching READY here only replaces the *repository-fetching and branch-selection* half of `cawaqs`’s own `make_Cawaqs.sh/make_Cawaqs_from_branches.sh` scripts. `ComplexGitSync` does not compile anything: the remaining `make -f Makefile` build step is still the user’s job, and only succeeds from this tree with `LIB_HYDROSYSTEM_PATH` left unset or pointed at the bootstrapped root — the alternative shared, decoupled install layout those scripts also support is out of scope for `ComplexGitSync`’s containment model.

`discover` (Level 3b below) is not a route into `cawaqs.cgs`: the 17 library repositories never coexist inside one directory tree until *after* a `.cgs` already lists them, so there is no single checkout for a filesystem walk to scan. Authoring this one from documented developer knowledge, the way Level 3a does for `cawaqsviz`, is the only available route.

5.5.3 Level 3 — cawaqsviz: three ways to the same .cgs

`cawaqsviz` (<https://gitlab.com/cawaqs/gviz/cawaqsviz>) is a real GitLab project, nested three path segments deep, with two GitHub children that were, until recently, plain git submodules. One of those children, `HydrologicalTwinAlphaSeries`, holds a submodule of its own at `docs/hydrological_twin`: four repositories on two levels, which is why every command below uses `-recursive`. It has no `.cgs` of its own anywhere upstream. That makes it a good vehicle for showing all three ways `ComplexGitSync` can produce a working `.cgs` for a project it does not already control the source of, ordered here by how much the operator has to already know about the tree before running the command. Full narrative, as one combined ten-step procedure rather than three independent routes, in `tutorials/03_adopting_a_real_project.md`.

3a — Write it by hand

The most explicit, and most error-prone, route: read the project’s own `.gitmodules` and topology, then author `examples/cawaqsviz.cgs` directly. Its corrected content and the exact mistakes the first draft made (wrong nested-subgroup identifier, missing `relative_path` overrides) are already spelled out in §5 “Local Authoring Spec”; this level is about running it, not re-deriving it:

```
$ pixi run cgitssync validate examples/cawaqsviz.cgs
$ pixi run cgitssync bootstrap examples/cawaqsviz.cgs cawaqsviz-demo
$ pixi run cgitssync view-tree --search-dir <path bootstrap printed>
```

`bootstrap` is used instead of `initialise` here because, unlike `CGSil1` above, this example is not meant to have `ComplexGitSync` cloned inside the tree it manages — see `bootstrap` in §5 for why the two commands pick different default locations.

3b — Derive it with `discover` (autodiscover mode)

The opposite route: start from nothing but a checkout, and let `ComplexGitSync` read the filesystem instead of writing the `.cgs` by hand. This is the route to prefer whenever a checkout is already available, since it cannot repeat the hand-authoring mistakes of 3a — `relative_path` falls out of the walk directly instead of being typed.

```
# A plain clone leaves submodule paths empty; initialise them first so
# discover has something to find at those paths. --recursive reaches the
# second level too: without it, hydrological_twin stays an empty
# directory and discover cannot see a repository that is not there.
$ git clone https://gitlab.com/cawaqs/gviz/cawaqsviz cawaqsviz
$ cd cawaqsviz && git submodule update --init --recursive

# Dry run: report what discover sees, write nothing.
$ pixi run cgitsync discover . --max-depth 3

# Satisfied with the report: draft the .cgs.
$ pixi run cgitsync discover . --write cawaqsviz-discovered.cgs
$ pixi run cgitsync validate cawaqsviz-discovered.cgs
```

The draft reconstructs 3a's hand-corrected file field-for-field: the root at `relative_path = "."` and every other repository at its real submodule path (`external/HydrologicalTwinAlphaSeries`, `docs/CWV_user_guide`, `external/HydrologicalTwinAlphaSeries/docs/hydrological_twin`) rather than their bare repo names. That last path lies inside `HydrologicalTwinAlphaSeries`, so the report marks it inside: `external/HydrologicalTwinAlphaSeries` and the file is read back with `HydrologicalTwinAlphaSeries` as its parent. No repository here has a `.cgs` of its own, and `discover` writes no `nested_config` override for that: the default "auto" already resolves a repository with zero nested `*.cgs` matches cleanly. See `discover` in §5 for exactly what is and is not reported (no network calls, unresolvable remotes are warned about and left out, never guessed at).

3c — Migrate it with `import-submodules`

The historical route: before either `.cgs` above existed, `cawaqsviz` tracked its children as real git submodules. This mode starts from that state and converts it, rather than describing a tree that is assumed to already be plain clones.

```
# Dry run against a real cawaqsviz checkout that still has submodules:
# reports name, path, URL, branch and declaring .gitmodules per
# submodule;
# changes nothing. --recursive reaches HydrologicalTwinAlphaSeries' own
# submodule; without it, only the root's two are converted.
$ pixi run cgitsync import-submodules /path/to/cawaqsviz --recursive

# Convert: drop the gitlinks, retire .gitmodules, extend .gitignore.
$ pixi run cgitsync import-submodules /path/to/cawaqsviz --recursive --
  apply

# Review and commit the resulting working-tree changes as usual. Two
# repositories have changes here, one per converted level.
$ cd /path/to/cawaqsviz && git status
```

```
$ git commit -m "chore: retire git submodules in favour of
  ComplexGitSync"
```

This is the whole job: turning gitlinks into plain clones on disk. It does not also write a `.cgs` — `.gitmodules` never records the root’s own identity, and a checkout worth converting already has a `.cgs` (3a) or can get one from `discover` (3b), run before or after `-apply`. The full before/after picture (index, `.gitmodules`, `.gitignore`) and the automated test that proves it against fixture repositories are in `tutorials/03_adopting_a_real_project.md` §4 — this level is the condensed command sequence, not a restatement of that walkthrough. `import-submodules` and `discover` answer different questions about the same tree: `discover` reports what is *checked out*, `import-submodules` reads what git *declares* in `.gitmodules` and acts on it — a submodule path that was never initialised is invisible to 3b but not to 3c.

A shared caveat across all three

`ComplexGitSync` stores no credentials and has no private-repository authentication story (§5 “`import-submodules`” and “`discover`”). All three routes above assume every repository in the tree is reachable anonymously or through credentials the ambient environment (`ssh-agent`, an HTTPS credential helper) already holds.

5.6 Configuration and Environment

`ComplexGitSync` uses project files for topology and does not require global topology configuration. Snapshot discovery can use `CGSHOME`; when it is not set, the `initialise(.cgs)` output-path default is `CGSPATH=../..` (nested mode: `ComplexGitSync` cloned inside the tree it manages); later commands can also walk upward from the current directory to find `.cgitsync/state/bootstrap(.cgs)` uses a different default instead — `CGSPATH=$HOME/.cgs/CGS<timestamp>/`, created automatically — since it is meant for a `ComplexGitSync` clone that is not nested inside any project tree; see `bootstrap` above. The logging and runtime-state subsystems use the user state directory by default:

- logs: `/.local/state/ComplexGitSync/logs/`
- latest runtime-state mapping: `/.local/state/ComplexGitSync/runtime/`

If `XDG_STATE_HOME` is set, these directories are rooted there instead.

5.7 Error Reference

ConfigValidationError Raised when `project` or `repos` is missing, a repository identifier is malformed, or a canonical field is invalid (for example an unknown `gitprovider`).

GitSyncError Raised when clone-time Git operations fail, for example because the remote branch cannot be resolved or the target directory is already occupied.

Exit code 1 General CLI failure (parse error, validation error).

Exit code 2 Command exists but is not yet implemented.

5.8 Troubleshooting

“`gitprovider_url` is required for custom provider” Add a `gitprovider_url` field to the offending `[[repos]]` entry in your `.cgs` file.

validate exits non-zero despite a correct-looking file Check that `project` is present and every `repos` entry uses `provider:owner/repository`. For advanced `[[repos]]` blocks without a `repository` shorthand, include `project_owner_name` and `project_name`.

Clone destination already exists and is not empty Re-run `clone` with `-target-dir` to force a clean location, or let `cgitsync` choose a suffixed default directory.

No cloneable branch found for <repo> Confirm that the remote exposes either the repo's `default_branch` or its `fallback_branch`.

PyYAML import error when calling `from_yaml/to_yaml` Add PyYAML in the Pixi environment: `pixi add pyyaml`.